

RADIO NAVIGATION PRACTICE SETS



MADE EASY BY CAPT P KUMAR

TOP CREW AVIATION (INDIA) 2023

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RADIO WAVE

1. In aviation, the reflection on ionosphere layers phenomenon is used in the following frequencies:

- A. VHF
- B. HF
- C. UHF
- D. VLF

2. Skip distance is the:

- A. Range from the transmitter to the first sky wave
- B. Highest critical frequency distance
- C. Wavelength distance of a certain frequency
- D. Thickness of the ionosphere

3. A radio signal loses strength as range from the transmitter increases, this is called:

- A. Propagation
- B. Refraction
- C. Attenuation
- D. Ducting

4. The skip zone of HF transmission will increase when the following change in circumstance occurs:

- A. Higher frequency and higher position of the reflecting ionospheric layer
- B. Lower frequency and higher position of the reflecting ionospheric layer
- C. Higher frequency and lower position of the reflecting ionospheric layer
- D. Lower frequency and lower position of the reflecting ionospheric layer

5. In the propagation of MF waves, the phenomenon of FADING is particularly found:

- A. At night, due to the combination of the sky and ground waves.
- B. By day, due to the combination of sky and ground waves.
- C. At night and when raining.
- D. By day and when raining.

6. What is the approximate maximum theoretical range at which an aircraft at FL130 could receive information from a VDF facility which is sited 1024 FT above MSL?

- A. 150 NM
- B. 220 NM
- C. 120 NM
- D. 180 NM

7. A radio wave is:

- A. An energy wave comprising an electrical field in the same plane as a magnetic field
- B. An electrical field alternating with a magnetic field
- C. An energy wave where there is an electrical field perpendicular to a magnetic field
- D. An energy field with an electrical component

8. The speed of radio waves is:

- A. 300 km per second
- B. 300 million meters per second
- C. 162 nm per second
- D. 162 million nm per second

9. The plane of polarization of an electromagnetic wave is:

- A. The plane of the magnetic field
- B. The plane of the electrical field
- C. The plane of the electrical or magnetic field dependent on the plane of the aerial
- D. None of the above

10. If the wavelength of a radio wave is 3.75 meters, the frequency is:

- A. 80 kHz
- B. 8 MHz
- C. 80 MHz
- D. 800 kHz

11. To carry out a phase comparison between two electromagnetic waves:

- A. Both waves must have the same amplitude
- B. Both waves must have the same frequency
- C. Both waves must have the same amplitude and frequency

D. Both waves must have the same phase

12. The process which causes the reduction in signal strength as range from a transmitter increases is known as:

- A. Absorption
- B. Diffraction
- C. Attenuation
- D. Ionisation

13. Which of the following will give the greatest surface wave range?

- A. 243 MHz
- B. 500 kHz
- C. 2182 kHz
- D. 15 MHz

14. a 300 KHz transmitter has an output of 1600 watts, the effective range over the sea will be:

- A. 52 nm
- B. 80 nm
- C. 35 nm
- D. 120 nm

15. What is the minimum height for an aircraft at a range of 200 nm to be detected by radar at 1600ft amsl?

- A. 25,500 ft
- B. 15,000 ft
- C. 40,000 ft
- D. 57,500 ft

16. Concerning HF communications, which of the following is correct?

- A. The frequency required in low latitudes is less than the frequency required in high latitudes.
- B. At night a higher frequency is required than by day.
- C. The frequency required is dependent on time of day but not the season.
- D. The frequency required for short ranges will be less than the frequency required for long ranges.

17. Which of the following statements concerning AM is correct?

- A. The amplitude of the RF is modified by the frequency of the AF



- B. The amplitude of the RF is modified by the amplitude of the AF
- C. The frequency of the RF is modified by the frequency of the AF
- D. The frequency of the RF is modified by the amplitude of the AF

18. Which of the following is an advantage of single sideband (SSB) emissions?

- A. More frequencies available
- B. Reduced power requirement
- C. Better signal/noise ratio
- D. All of the above

19. Which of the following statements concerning FM is correct?

- A. The amplitude of the RF is modified by the frequency of the AF
- B. The amplitude of the RF is modified by the amplitude of the AF
- C. The frequency of the RF is modified by the frequency of the AF
- D. The frequency of the RF is modified by the amplitude of the AF

20. Radio waves travel through the atmosphere

- A. At a speed which varies in direct relation to power of the transmission
- B. At a speed which varies directly as the frequency
- C. At a constant speed
- D. None of the above

21. The skip distance

- A. The difference between the aural null and maximum reflection
- B. The distance between the crests to the crest of the radio wave
- C. The distance between the point where the ground wave fades out and reflection sky wave strike the earth.
- D. The distance between Tx and reflected sky wave initially strikes the earth.

22. An advantage of VHF compared to low frequency radio is

- A. Speed reception as the waves travel faster
- B. Reduction in static interference
- C. Much greater range is possible
- D. None of the above

23. The best frequency band to use in long distance communication is:

- A. Medium
- B. High
- C. Low

D. None of the above

24. Electron density

- A. In E (Kennelly heavy side) layer is greater than F layer
- B. In E layer is less than F layer
- C. Is same in both the layer
- D. None of the above

25. Radio waves travelling from sea to land

- A. Bends towards normal
- B. Bends away from the normal
- C. Does not bend at all
- D. None of the above

26. If a radio wave is describe as horizontally polarized, in what planes are the i.) Magnetic and ii.) Electrical fields oriented?

- A. a)i) horizontal ii) horizontal
- B. b) i) horizontal ii) vertical
- C. c)i) vertical ii) horizontal
- D. d)i) horizontal ii) circular

27. Which emission designation would apply to single channel amplitude modulated single sideband signal with suppressed carrier wave carrying analogue information from aural (telephonic) reception?

- A. A2A
- B. A8W
- C. J3E
- D. PON

28. What is a hertz?

- A. The wavelength in cycles second
- B. A frequency of one cycle per second
- C. The wavelength corresponding to one cycle per second
- D. The frequency in cycles per second

29. An AM signal at frequency f_m carried on a transmitted frequency f_c produces:

- A. One sideband of transmission at $f_c + 2f_m$
- B. Two sidebands of transmission at $f_c + f_m$ and $f_c - f_m$

- C. One sideband of transmission at $f_s - 2f_m$
- D. Two sidebands of transmission at $f_c + 2f_m$ and $f_c - 2f_m$

30. What happens to the direction and speed of propagation of radio waves as they pass through air whose density is increasing?

- A. The waves bend towards the high density and speed up
- B. The waves bend towards the high density and speeds down
- C. The waves bend away from the high density and speed up
- D. None of the above

31. What happens in practical terms to the layers of the ionosphere at night?

- A. The D layer thickens and E layer rises
- B. The D layer thins and E layer rises
- C. The E layer thickens and D layer rises
- D. The E layer thins and D layer rises

32. If the transmission power of a signal sent by both the surface wave and sky wave is doubled, what will happen to the dead space?

- A. Nothing
- B. The dead space will reduce from both the ends
- C. The dead space will reduce as skip distance reduces
- D. The dead space will reduce from the transmission end

33. If the transmission frequency of a signal sent by both the surface wave and sky wave is doubled, what will happen to the dead space?

- A. Nothing
- B. The dead space will increase from both the ends
- C. The dead space will reduce as skip distance reduces
- D. The dead space will reduce from the transmission end

34. A single sideband transmission:

- A. Normally transmits the carrier wave also?
- B. Is used to reduce the atmospheric noise received
- C. Has an emission designated of MJ?
- D. Is usually in SHF band?

35. Which wave is used for VHF communications?

- A. Sky wave
- B. Surface wave

- C. Ground wave
- D. Space wave

36. UHF freq band has wavelength between

- A. 1 to 10 m
- B. 10 to 100 m
- C. 10 cm to 100 cm
- D. 1 cm to 10 cm

37. You are at an altitude of 9,000 feet. At a range of 200 nm from a VHF communications transmitter and you are receiving a good signal.

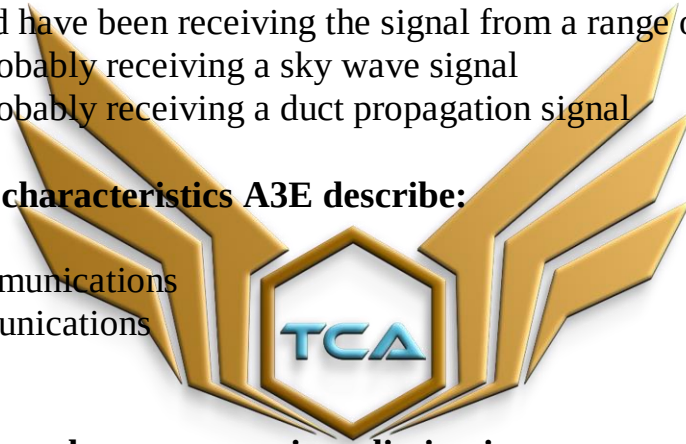
- A. You should expect this since the transmitter is at a height of 2,000 feet
- B. You should have been receiving the signal from a range of 240 nm
- C. You are probably receiving a sky wave signal
- D. You are probably receiving a duct propagation signal

38. The emission characteristics A3E describe:

- A. ILS
- B. VHF communications
- C. HF communications
- D. VOR

39. Refraction of an electro-magnetic radiation is:

- A. The bending of its propagation path as it passes through or over areas of different electrical conductivity
- B. The loss of power as it passes through or over areas of different electrical conductivity
- C. Is bending resultant from reflection from objects
- D. Is loss of power through reflection from objects



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VHF DIRECTION FINDING (VDF)

- 1. When conducting a QGH approach responsibility for interpreting the procedure rests with _ and on a VDF approach responsibility rests with**
 - A. the controller; the controller
 - B. the pilot; the pilot
 - C. the pilot; the controller
 - D. the controller; the pilot

- 2. In which one of the following circumstances is ground direction finding (VDF) likely to be used to fix an aircraft's position?**
 - A. When contacting ATC to join controlled airspace from the open FIR
 - B. When declaring an emergency on any frequency
 - C. When using the emergency VHF frequency 121.5 MHz
 - D. On first contact with ATC on crossing an international FIR boundary

- 3. Which of the following does NOT affect the accuracy of VDF bearings:**
 - A. sky waves
 - B. duct propagation
 - C. ground reflections
 - D. synchronous transmissions

- 4. What is QTE?**
 - A. Magnetic track from the station
 - B. Magnetic track to the station
 - C. True track from station
 - D. True track to the station

- 5. What is the "Q" code for a magnetic bearing from a VDF station?**
 - A. "Request QNH"
 - B. "Request QDR"
 - C. "Request QTE"
 - D. "Request QDM"

- 6. Which Q code would give a magnetic bearing from a VDF station?**
 - A. QDM

- B. QDR
- C. QUJ
- D. QTE

7. The VDF Homer service provides:

- A. Accurate bearings on demand
- B. A QGH procedure to transiting aeroplanes
- C. Bearings which will normally be within $\pm 5^\circ$ accuracy to a range of 200NM
- D. Bearings which may be affected by synchronous transmissions

8. A class C magnetic bearing is received from a station. This is:

- A. A QUJ accurate to $\pm 5^\circ$
- B. A QDM accurate to $\pm 10^\circ$
- C. A QTE accurate to $\pm 5^\circ$
- D. A QDR accurate to $\pm 10^\circ$

9. The VHF direction finder uses the following wavelengths:

- A. decimetric.
- B. hectometric.
- C. metric.
- D. centimetric.

10. Which of the following is an advantage of Ground/DF (VDF) let-down?

- A. It only requires a VHF radio to be fitted to the aircraft
- B. It is pilot interpreted and does not require the assistance of ATC
- C. It does not require any special equipment to be fitted to the aircraft
- D. It does not require any special equipment, apart from a VHF radio, to be installed in the aircraft or on the ground

11. A VDF QDM given without an accuracy classification may be assumed to be accurate to within:

- A. 2 degrees
- B. 5 degrees
- C. 7.5 degrees
- D. 10 degrees

12. An aircraft is passed a true bearing from a VDF station of 353° . If variation is 8°E and the bearing is classified as 'B' then the:

- A. QDM is $345^{\circ} \pm 5^{\circ}$
- B. QDR is $345^{\circ} \pm 2^{\circ}$
- C. QTE is $353^{\circ} \pm 5^{\circ}$
- D. QUJ is $353^{\circ} \pm 2^{\circ}$

13. A VDF Bearing can be obtained:

- A. Only on a frequency of 121.5 MHz
- B. On the approach frequency
- C. On the frequency notified for VDF services
- D. On the frequency agreed between the pilot and ATC but chosen from one of the available ATC frequencies



NDB & ADF

1. The Automatic Direction Finder uses the following wavelengths:
 - A. centimetric.
 - B. metric.
 - C. decimetric.
 - D. hectometric or kilometric.

2. An aircraft is ""homing"" to a radio beacon whilst maintaining a relative bearing of zero. If the magnetic heading decreases, the aircraft is experiencing :
 - A. zero drift
 - B. left drift
 - C. right drift
 - D. a wind from the west

3. An RMI indicates aircraft heading. To convert the RMI bearings of NDBs and VORs to true bearings the correct combination for the application of magnetic variation is:
 - A. NDB: aircraft position VOR: beacon position
 - B. NDB: beacon position VOR: beacon position
 - C. NDB: beacon position VOR: aircraft position
 - D. NDB: aircraft position VOR: aircraft position

4. A radio beacon has an operational range of 10 NM. By what factor should the transmitter power be increased in order to achieve an operational range of 20 NM?
 - A. Six
 - B. Four
 - C. Eight
 - D. Two

5. 'Night Effect' which causes loss of signal and fading, resulting in bearing errors from NDB transmissions, is due to:
 - A. static activity increasing at night particularly in the lower frequency band

- B. interference from other transmissions and is maximum at dusk when east of the NDB
 - C. sky wave distortion of the null position and is maximum at dawn and dusk
 - D. the effect of the Aurora Borealis
- 6. Errors caused by the effect of coastal refraction on bearings at lower altitudes are maximum when the NDB is:**
- A. inland and the bearing crosses the coast at an acute angle
 - B. near the coast and the bearing crosses the coast at right angles
 - C. inland and the bearing crosses the coast at right angles
 - D. near the coast and the bearing crosses the coast at an acute angle
- 7. There are two NDBs, one 20 NM inland, and the other 50 NM inland from the coast. Assuming that the error caused by coastal refraction is the same for both propagations, the extent of the error in a position line plotted by an aircraft that is over water will be:**
- A. the same from both beacons when the aircraft is on a relative bearing of 180° and 360°
 - B. greater from the beacon that is 50 NM inland
 - C. greater from the beacon that is 20 NM inland
 - D. the same from both beacons when the aircraft is on a relative bearing of 090° and 270°
- 8. Which of the following is likely to have the greatest effect on ADF accuracy?**
- A. Interference from other NDBs, particularly during the day
 - B. Frequency drift at the ground station
 - C. Interference from other NDBs, particularly at night
 - D. Mutual interference between aircraft aerials
- 9. ADF bearings by an aeroplane by day within the published protection range should be accurate to within a maximum error of:**
- A. $\pm 5^\circ$
 - B. $\pm 10^\circ$
 - C. $\pm 2.5^\circ$
 - D. $\pm 2^\circ$

10. Which of the following is the ICAO allocated frequency band for ADF receivers?

- A. 200 - 1750 kHz
- B. 255 - 455 kHz
- C. 300 - 3000 kHz
- D. 200 - 2000 kHz

11. In order to obtain an ADF bearing the:

- A. BFO switch must be selected to 'ON'
- B. sense aerial must be tuned separately
- C. mode selector should be switched to 'loop'
- D. signal must be received by both the sense and loop aerials

12. Factors liable to affect most NDB/ADF system performance and reliability include:

- A. static interference - night effect - absence of failure warning system
- B. static interference - station interference - latitude error
- C. height error - station interference - mountain effect
- D. coastal refraction - lane slip - mountain effect

13. Which one of the following disturbances is most likely to cause the greatest inaccuracy in ADF bearings?

- A. Quadrantal error
- B. Coastal effect
- C. Local thunderstorm activity
- D. Precipitation interference

14. The phenomenon of coastal refraction which affects the accuracy of ADF bearings:

- A. During night
- B. Can be minimized by using beacons situated well inland
- C. Can be minimized by taking bearing where the signal crosses the coastline at right angles.
- D. Right angles.

15. An aircraft is intending to track from NDB 'A' to NDB 'B' on a track of 050° (T), heading 060° (T). if the RBI shows the relative bearing of 'A' to

be 180° and the relative bearing of 'B' to be 330° then the aircraft is:

- A. Port of track and nearer 'A'
- B. Port of track and nearer 'B'
- C. Starboard of track and nearer 'A'
- D. Starboard of track and nearer 'B'

16. In order to tune, identify and monitor NON A1A NDB emission the BFO should be used as follows:

Tune	Identify	Monitor
A. On	On	off
B. On	On	On
C. On	Off	Off
D. Off	Off	Off

17. The magnitude of the error in position lines derived from ADF bearing that are affected by coastal refraction may be reduced by:

- A. Selecting beacons situated well inland.
- B. Only using beacons within the designated operational coverage
- C. Choosing NON A2A beacons
- D. Choosing beacons on or near the coast

18. The phenomena of coastal refraction affecting ADF bearing is caused by the signal When it reaches the coastline and bending..... the normal to the coast:

- A. Accelerating, towards
- B. Decelerating, towards
- C. Accelerating, away from
- D. Decelerating, away from

19. In an ADF system, night effect is most pronounced:

- A. During long winter nights
- B. When the aircraft is at low altitude
- C. When the aircraft is at high altitude
- D. At dusk and dawn

20. When the induced signal from the loop and the sense antenna is combined in an ADF receiver, the resultant polar diagram is:

- A. A limacone
- B. A cardioid
- C. Figure of eight shaped
- D. Circular

21. When flying over the sea and using an inland NDB to fixed position with a series of position lines the plotted position, in relation to the aircraft's actual position, will be:

- A. Further from the coast
- B. Closer to the coast
- C. Co-incident
- D. Inaccurate due to the transmitted wave front decelerating

22. An aircraft on a heading of 235°(M) shows an RMI reading of 090 with respect to an NDB. Any quadrantal error which is affecting the accuracy of this bearing is likely to be:

- A. A maximum value
- B. A very small value
- C. Zero, since quadrantal error affects only the RBI
- D. Zero, since quadrantal error affects only the VOR

23. The principle propagation path employed in an NDB/ADF system is:

- A. Sky wave
- B. Surface wave
- C. Direct wave
- D. Ducted wave

24. The ADF of an aircraft on a heading of 189° (T) will experience the greatest effect due to quadrantal error if the NDB bears:

- A. 234°(T)
- B. 279°(T)
- C. 225°(T)
- D. 145°(T)

25. Aircraft enroute place A to B reads on ADF relative bearing 188 on tracking out from A and 008 While tracking in to place B.

- A. a/c is on track
- B. a/c is port of track
- C. a/c is starboard of track
- D. a/c is on single track

26. Total distance = 135 nm, ADF reads 170° at 90 nm. Find correction to be applied to destination.

- A. 30° port
- B. 30° starboard
- C. 15° port
- D. 15° starboard

27. An ac on outbound track Delhi is on a heading to compensate 10° S drift its ADF tuned to Delhi NDB should read:

- A. 190°
- B. 010°
- C. 180°
- D. 170°

28. An ac is on track 300 and heading 300 from A to B total distance 135 nms. After 90 NMs from A, the aircraft ADF turned to NDB of place A reads 190. The a/c distance off track and heading correction to place B are:

- A. 15 nms and 30 S
- B. 30 nms and 30 S
- C. 15 nms and 30 P
- D. 35 nms and 30 p

29. An a/c maintaining track, is on radial 105 with heading 120 (T) with var 8° E. find relative bearing indicated on ADF.

- A. 183°
- B. 173°
- C. 187°
- D. 168°

30. Wingtip bearing change = 10° elapsed time between bearing = 8 min rate of fuel consumption = 11.8 GPH The fuel required to fly to the station

- A. 8.2 gal
- B. 9.4 gal
- C. 15.7 gal
- D. 48.4 gal

31. Determine approx. time and distance to NDB stn if 5° bearing change occurs in 1.5 min with a TAS of 95 kts.

- A. 11 min and 40 nm
- B. 16 min and 14.3 nm
- C. 18 min and 28.5 nm
- D. 18 min and 33 nm

32. The rel. bearing of station changes from 270 to 2.5 min the time to station is min

- A. 6
- B. 12
- C. 9
- D. 15

33. ADF Quadrantile error is caused by:

- A. Static build up on the airframe and St. Elmo's fire.
- B. The aircraft's major electrical axis, the fuselage, reflecting and re-radiating the incoming NDB transmissions
- C. Station interference and/or night effect.
- D. NDB signals speeding up and bending as they cross from a land to water propagation path

34. The magnitude of the error in position lines derived from ADF bearing that are affected by coastal refraction may be reduced by:

- A. Selecting beacons situated well inland.
- B. Only using beacons within the designated operational coverage
- C. Choosing NON A2A beacons
- D. Choosing beacons on or near the coast

35. The BFO facility on ADF equipment should be used as follows when an NDB having NON A1A type emission is to be used:

- A. BFO on for tuning and identification but may be turned off for monitoring
- B. BFO on for turning but can be turned off for monitoring and identification purpose
- C. BFO off during turning, identification and monitoring because this type of emission is not modulated
- D. BFO should be switched on for turning, indent and monitoring

36. In order to resolve the 180° directional ambiguity of a directional LOOP aerial its polar diagram is combined with that of a SENSE aerial..... To produce a whose single null ensures the ADF needle moves the shortest distance to indicate the correct.....

- A. At the aircraft, cardioid, radial
- B. At the transmitter, limacine, bearing
- C. At the aircraft, limacine, bearing
- D. At the aircraft, cardioid, bearing

37. In an ADF system, night effect is most pronounced:

- A. During long winter nights
- B. When the aircraft is at low altitude
- C. When the aircraft is at high altitude
- D. At dusk and dawn



VOR

1. Given: VOR station position $N61^{\circ} E025^{\circ}$, variation $13^{\circ}E$. Estimated position of an aircraft $N59^{\circ} E025^{\circ}$, variation $20^{\circ}E$ What VOR radial is the aircraft on?

- A. 167°
- B. 347°
- C. 160°
- D. 193°

2. A VOR is sited at position $58^{\circ} 00' N 073^{\circ} 00' W$ where the magnetic variation equals $32^{\circ}W$. An aircraft is located at position $56^{\circ} 00' N 073^{\circ} 00' W$ where the magnetic variation equals $28^{\circ}W$. The aircraft is on VOR radial:

- A. 360
- B. 208
- C. 212
- D. 180

3. An airway 10 NM wide is to be defined by two VORs each having a resultant bearing accuracy of plus or minus 5.5° . In order to ensure accurate track guidance within the airway limits the maximum distance apart for the transmitter is approximately:

- A. 165 NM
- B. 50 NM
- C. 105 NM
- D. 210 NM

4. What is the theoretical maximum range that an aircraft at flight level 360 will obtain from a VOR beacon situated at 900 feet above mean sea level?

- A. 274 nms
- B. 255 nms
- C. 112 nms
- D. 224nms

5. A conventional VOR;

- A. Has an FM reference signal and an AM variable signal
- B. Has a 150 Hz signal and a 90 Hz variable signal
- C. Has an AM reference signal and a 150 Hz variable signal
- D. Has an AM reference signal and an FM variable signal.

6. The OBS on a deviation indicator is set 330° and gives a 3 dots fly right demand with FROM indicated. What is the QDM of the aircraft to the station?

- A. 144
- B. 324
- C. 336
- D. 156

7. An aircraft is homing towards a VOR which marks the center line of an airway. The beacon is 100 nms distant. If the pilot had the airway QDM set on the OBS what deflection of the deviation indication would be given if the aircraft was on the boundary of the airway?

Assume that one dot equals 2 degrees.

- A. 3 dots
- B. 2 dots
- C. 2.5 dots
- D. 1.5 dots

8. What is the theoretical maximum range that an aircraft at flight level 420 will obtain from a VOR beacon situated at 400 feet above mean sea level?

- A. 225nm
- B. 256nm
- C. 281nm
- D. 257nm

9. Concerning conventional and Doppler VORs (DVOR), which of the following is correct?

- A. There is no way of knowing from the instrumentation display which type is being used
- B. The DVOR will always have a “D” in the indent
- C. The DVOR has a higher pitch indent than the standard VOR
- D. The conventional VOR has less site error

10. In a Doppler VOR (DVOR) the reference signal is the bearing signal is and the direction of rotation of the bearing signal is

- A. AM, FM, anti-clockwise
- B. AM, FM, clockwise
- C. FM, AM, anti-clockwise
- D. FM, AM, clockwise

11. An aircraft is attempting to home to a VOR on the 064 radial. The CDI shows 4 dots fly right with a TO indication. At the same time the co-located DME shows a range of 45 nm. Where is the aircraft in relation to the required track?

- A. 6 nm right of track
- B. 3 nm right of track
- C. 6 nm left of track
- D. 3 nm left of track

12. A VOR beacon ceases to transmit its normal identification which is substituted by ‘TST’ this means that:

- A. The beacon may be used providing that extreme caution is used.
- B. The beacon is undergoing maintenance or calibration and should not be used.
- C. This is a temporary shorts range
- D. The beacon is under test and pilot using it should report its accuracy to aircraft control.

13. What is the approximate maximum range that an aircraft flying at 25000’ would expect to obtain from a VOR beacon situated 900’ above mean sea level?

- A. 220 nm
- B. 100 nm
- C. 235 nm
- D. 198 nm

14. An aircraft is on the airway boundary range 100 nm from a VOR marking the airway center line. Assuming that each dot equates to 2° how many dots deviation will be shows on the deviation indicator?

- A. 3.0 dots
- B. 2.5 dots
- C. 2.0 dots
- D. 1.5 dots

15. An aircraft is required to intercept and home to a VOR along the 064 radial. This OBS should be set to:

- A. 064 to get correct needle sense and a TO indication
- B. 244 to get correct needle sense and a TO indication
- C. 064 to get correct needle sense and a FROM indication
- D. 244 to get correct needle sense and a FROM indication

16. The type of emission radiated by a VOR beacon is:

- A. A double channel VHF carrier with one channel being amplitude modulated and the second channel being frequency modulated.
- B. Single channel VHF carrier wave amplitude modulated at 30 Hz with a sub carrier being frequency modulated at 30 Hz.
- C. A VHF carrier wave with a 90 Hz frequency modulated and a 150 Hz amplitude modulation
- D. A VHF pulse modulated emission with a pulse repetition frequency of 30 pps.

17. An aircraft wishes to track towards a VOR along the 274 radial. If variation is 10° W what should be set on the OBS?

- A. 274
- B. 264

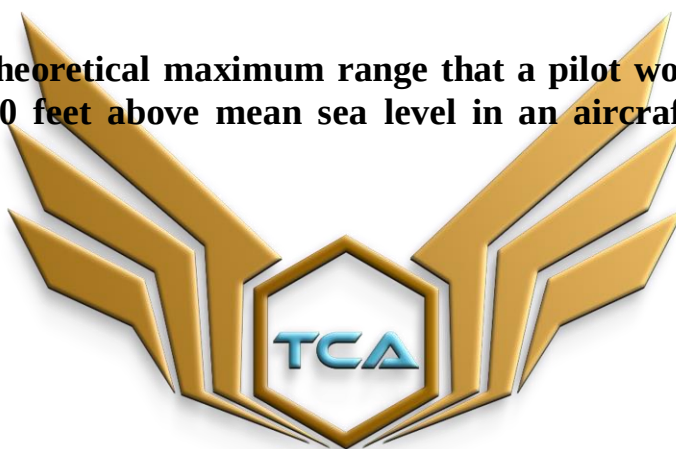
- C. 094
- D. 084

18. An aircraft is tracking away from a VOR on a heading of 287°M with 14° starboard drift. If the variation is 6°W what is the phase difference between the reference and variable phase components of the VOR transmission?

- A. 121°
- B. 295
- C. 301°
- D. 315°

19. What is the theoretical maximum range that a pilot would obtain from a VOR situated 900 feet above mean sea level in an aircraft flying at 18,000 feet?

- A. 168 nm
- B. 188 nm
- C. 205 nm
- D. 250 nm



20. An aircraft is attempting to home to a VOR beacon. The pilot has set 329 on the OBS of the deviation indicator. If the aircraft is situated on the 152 radial then the deviation indicator will show

- A. One and a half dots fly right
- B. One and a half dots fly left.
- C. Three dots fly right.
- D. Three dots fly left

21. A VOR receiver in an aircraft measures the phase difference from a DVOR as 220° . Which radial is the aircraft on?

- A. 140
- B. 040
- C. 320
- D. 220

22. The RMI indicates the aircraft magnetic heading. To convert the RMI bearings of NDBs and VORs to true bearings, the correct combination for the application of magnetic variation is:

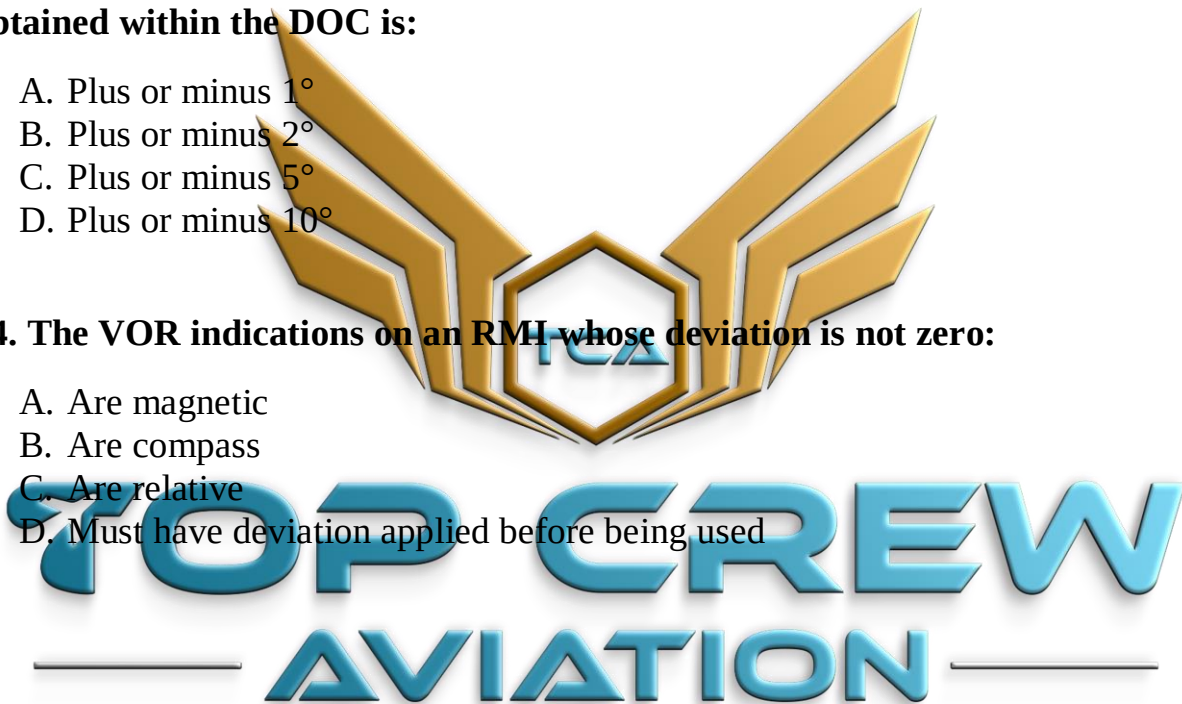
NDB	VOR
A. Beacon position	aircraft position
B. Beacon position	beacon position
C. Aircraft position	beacon position
D. Aircraft position	aircraft position

23. The normal maximum error which might be expected with a VOR bearing obtained within the DOC is:

- A. Plus or minus 1°
- B. Plus or minus 2°
- C. Plus or minus 5°
- D. Plus or minus 10°

24. The VOR indications on an RMI whose deviation is not zero:

- A. Are magnetic
- B. Are compass
- C. Are relative
- D. Must have deviation applied before being used



ILS

1. **The OUTER MARKER of an Instrument Landing System (ILS) facility transmits on a frequency of:**
A. 300 MHz and is modulated by Morse at two dashes per second
B. 200 MHz and is modulated by alternate dot/dash in Morse
C. 75 MHz and is modulated by alternate dot/dash in Morse
D. 75 MHz and is modulated by Morse at two dashes per second
2. **An aircraft carrying out a 3° glidepath ILS approach experiences a reduction in ground speed from 150 kt at the outer marker to 120 kt over the threshold. The effect of this change in ground speed on the aircraft's rate of descent will be a decrease of approximately:**
A. 150 FT/MIN
B. 250 FT/MIN
C. 50 FT/MIN
D. 100 FT/MIN
3. **An aircraft carrying out an ILS approach is receiving more 90 Hz than 150 Hz modulation notes from both the localiser and glide path transmitters. The ILS indication will show:**
A. Fly right and fly down
B. Fly left and fly down
C. Fly right and fly up
D. Fly left and fly up
4. **Using the 1 in 60 rule calculate the height on a 3° glide path of an aircraft 4.5 NM from touchdown.**
A. 1480 ft
B. 1420 ft
C. 1370 ft
D. 1230 ft
5. **The visual and aural indications of the ILS outer marker are:**
A. A blue light and 2 dashes per second of a 1300 Hz modulated tone
B. An amber light and alternate dots and dashes of a 1300 Hz modulated tone

- C.A white light and 6 dots per second of a 30 Hz modulated tone
- D.A blue light and 2 dashes per second of 400 Hz modulated tone

6. What approximate rate of descent is required in order to maintain a 3° glide path at a ground speed of 120 kt?

- A.550 FT/MIN
- B.600 FT/MIN
- C.800 FT/MIN
- D.950 FT/MIN

7. The sensitive area of an ILS is the area aircraft may not enter when:

- A.ILS operations are in progress
- B.category 1 ILS operations are in progress
- C.category II/III ILS operations are in progress
- D.the ILS is undergoing calibration

8. ILS is subject to false glide paths resulting from:

- A.back-scattering of antennas
- B.spurious signals reflected by nearby obstacles
- C.multiple lobes of radiation patterns in the vertical plane
- D.ground returns ahead of the antennas

9. For a category one ILS glide path of 3.3° the coverage is:

- A.1.49 to 5.77 degrees
- B.1.49 to 5.94 degrees
- C.1.65 to 5.77 degrees
- D.1.65 to 5.94 degrees

10.What is the colour sequence when passing over an Outer, Middle and Inner Marker beacon?

- A.white,amber,blue
- B.amber,white,green
- C.blue,amber,white
- D.blue,green,white

11.The coverage of the ILS glide slope with respect to the localiser centreline is:

- A - +/- 10° to 8 nm

- B - +/- 10° to 25 nm
- C - +/- 8° to 10 nm
- D - +/- 35° to 17 nm

12. The reason why pre take-off holding areas are sometimes further from the active runway when ILS Category 2 and 3 landing procedures are in progress than during good weather operations is:

- A. heavy precipitation may disturb guidance signals
- B. aircraft manoeuvring near the runway may disturb guidance signals
- C. to increase distance from the runway during offset approach operations
- D. to increase aircraft separation in very reduced visibility conditions

13. The coverage of the ILS localiser at 17 nm is guaranteed up to an angle either side of the extended centreline of:

- A. 30 degrees
- B. 25 degrees
- C. 35 degrees
- D. 10 degrees

14. In which frequency band does an ILS glide slope transmit?

- A. VHF
- B. UHF
- C. SHF
- D. EHF

15. An aircraft on an ILS approach is receiving more 90 Hz modulation than 150 Hz modulation in both localiser and glide path. The correct action to regain the centreline and glide path would be to:

- A. increase rate of descent and fly left
- B. reduce rate of descent and fly right
- C. increase rate of descent and fly right
- D. reduce rate of descent and fly left

16. What is the approximate angular coverage of reliable navigation information for a 3° ILS glide path out to a distance of 10 NM?

- A. 0.45° above the horizontal to 1.75° above the glide path and 8° each side of the localiser centreline
- B. 1.35° above the horizontal to 5.25° above the horizontal and 8° each side of the localiser centreline

- C. 0.7° above and below the glide path and 2.5° each side of the localiser centreline
D. 3° above and below the glide path and 10° each side of the localiser centreline

17. Where, in relation to the runway, is the ILS localiser transmitting aerial normally situated?

- A. On the non-approach end of the runway about 300m from the runway on the extended centreline
B. At the approach end of the runway about 300m from touchdown on the centreline
C. At the non-approach end about 150 m to one side of the runway and 300m along the extended centreline
D. At the approach end about 150m to one side of the runway and 300m from touchdown

18. An aircraft tracking to intercept the Instrument Landing System (ILS) localiser inbound on the approach side, outside the published ILS coverage angle:

- A. will receive signals without identification coding
B. will not normally receive signals
C. may receive false course indications
D. can expect signals to give correct indications

19. Which of the following is an ILS localiser frequency?

- A. 112.10 MHz
B. 108.25 MHz
C. 110.20 MHz
D. 109.15 MHz

20. The outer marker of an ILS with a 3° glide slope is elevated 4.6 NM from the threshold. Assuming a glide slope height of 50 FT above the threshold, the approximate height of an aircraft passing the outer marker is:

- A. 1,400 FT
B. 1,450 FT
C. 1,350 FT
D. 1,300 FT

21. The principle of operation of an ILS localiser transmitter is based on two overlapping lobes that are transmitted on (i) frequencies and carry different (ii).

- A.(i) the same; (ii) modulation frequencies
- B.(i) the same; (ii) phases
- C.(i) different; (ii) modulation frequencies
- D.(i) different; (ii) phases

22. A Cat III ILS glide path transmitter provides reliable guidance information down to:

- A. the surface of the runway
- B. a maximum height of 200 ft above the runway
- C. a maximum height of 100 ft above the runway
- D. a maximum height of 50 ft above the runway

23. What frequency is assigned to all ILS marker beacons?

- A. One chosen from between 108-112 MHz at odd tenths
- B. 75 MHz
- C. 90 Hz
- D. 150 Hz

24. Which of the following statements is true, in respect of an ILS?

- A. If the glide path is not operating, the ILS will be switched off
- B. An ILS cannot be used if either of the outer or middle markers is switched off
- C. The glide path frequency is paired with the marker frequency
- D. The glide path transmits on UHF

25. Which of the following statements is TRUE?

- A. A localiser back beam should only be used for approaches if there is a published procedure
- B. All localisers have back beams. They provide guidance in the event of a missed approach
- C. Localiser back beams are never checked for accuracy
- D. A localiser back beam will always provide reversed steering signals

26. An ILS category II ground installation is one that is capable of providing guidance to a height of:

- A.15m above the horizontal plane containing the threshold
- B.60m above the horizontal plane containing the threshold
- C.15m on QNH
- D.60m on QNH

27.A Category 1 Instrument Landing System (ILS) ground installation provides accurate guidance from coverage limit down to:

- A.runway surface
- B.200 feet above the inner marker
- C.200 feet above the runway threshold
- D.50 feet above ILS reference point

28.According to ICAO 8168, what is regarded as the maximum safe deviation below the glide path during ILS approach?

- A.Half scale deflection
- B.One quarter scale deflection
- C.Three quarter scale deflection
- D.Full scale deflection

29.Full deflection on a glide slope indicator indicates that the aircraft is:

- A.2.5° above or below the correct glide path
- B.0.7° above or below the correct glide path
- C.0.5° above or below the correct glide path
- D.1.25° above or below the correct glide path

30.On what carrier frequency does the inner marker transmit?

- A.Same frequency as the localiser
- B.75 MHz
- C.Same frequency as the glide path
- D.3000 Hz

31.What is measured in order to establish aircraft position in relation to the localiser beam on an ILS?

- A.The difference in phase between the 90 Hz modulation and the 150 Hz modulation
- B.The difference in depth between the 90 Hz modulation and the 150 Hz modulation
- C.The bearing to the localiser antenna found by means of a loop antenna

D.The difference in time between the 90 Hz modulation and the 150 Hz modulation

32.Which of the following alternatives is correct regarding audio and visual signals in cockpit when passing overhead a middle marker?

- A.Audio: 400 Hz, 2 dashes per second. Visual: Blue light flashes
- B.Audio: 1300 Hz, alternating dots and dashes. Visual: Amber light flashes
- C.Audio: 75 MHz, 2 dashes per second. Visual: Blue light flashes
- D.Audio: 3000 Hz, alternating dots and dashes. Visual: Amber light flashes

33.Which of the following elements of an ILS transmit in the VHF band?

- A.Localiser only
- B.Marker beacons only
- C.Glide path and marker beacons
- D.Localiser and marker beacons

34.ILS marker beacons do not interfere with each other because:

- A.They operate on different modulations
- B.They operate at different frequencies
- C.They transmit in narrow vertical beams
- D.They transmit low power signals, which cannot be detected by the aeroplane's receiver

35.ILS markers are identified in the aeroplane by colour light and audio signal. The identification of the outer marker is:

- A.High-pitched dashes; amber light
- B.Low-pitched dots and dashes; amber light
- C.High-pitched dots and dashes; blue light
- D.Low-pitched dashes; blue light

36.The azimuth and area coverage of a Cat I ILS localiser is:

- A.8° at 10 nm, 25° at 25 nm
- B.35° at 17 nm, 10° at 25 nm
- C.8° at 35 nm, 25° at 10 nm
- D.8° at 10 nm, 10° at 25 nm

37.ILS back beams may be received:

- A.When flying outside the area of coverage
- B.Never

- C. When approaching the ILS from behind the glide path aerial
- D. When approaching the ILS from behind the localiser aerial

38. The emission characteristics of the ILS and a typical localiser frequency are:

- A. A9W 329.30 MHz
- B. A8W 110.30 MHz
- C. A9W 110.70 MHz
- D. A8W 113.30 MHz

39. An aircraft is flying downwind outside the coverage of the ILS. The CDI indications will be:

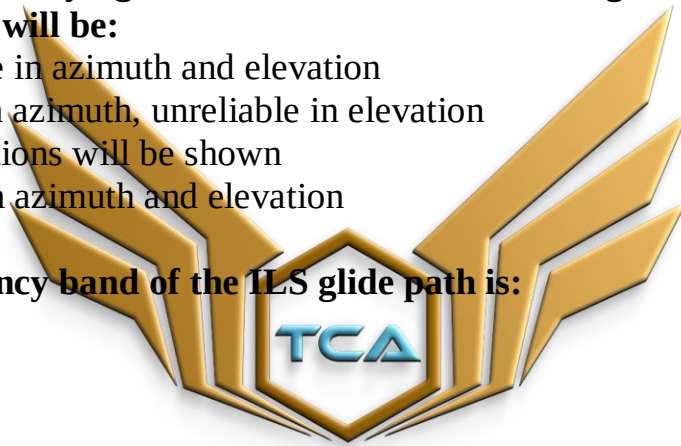
- A. unreliable in azimuth and elevation
- B. reliable in azimuth, unreliable in elevation
- C. no indications will be shown
- D. reliable in azimuth and elevation

40. The frequency band of the ILS glide path is:

- A. UHF
- B. VHF
- C. SHF
- D. VLF

41. The minima for a CAT II ILS are:

- A. Height: 100 ft ; RVR: 300 m
- B. Height: 100 ft ; RVR: 400 m
- C. Height: 50 ft ; RVR: 300 m
- D. Height: 50 ft ; RVR: 400 m



**TOP CREW
AVIATION**

RADAR

1. With regards to radio waves, which statement is true?

- A. They are reflected by metallic objects with a size compatible to the wavelength
- B. The longer the wavelength the greater the surface attenuation
- C. They travel at 186,000 nm a second in a vacuum
- D. High frequencies need large aerials

2. The main factor which affects the maximum range of a pulse radar is:

- A. the pulse repetition frequency
- B. the size of the radar screen
- C. the frequency of the radar transmission
- D. the aerial system size

3. In relation to radar systems that use pulse technology, the term Pulse Recurrence Rate (PRR) signifies the:

- A. ratio of pulse period to pulse width
- B. delay after which the process re-starts
- C. the number of cycles per second
- D. number of pulses per second

4. The advantages of CW radar systems over pulse radar systems are:

- A. they are more reliable
- B. the transmitter/receiver aerial system is smaller and less complex
- C. there is no minimum range
- D. they offer better long range performance

5. The main advantage of a slotted scanner is:

- A. reduces side lobes and directs more energy into the main beam
- B. removes the need for azimuth slaving
- C. side lobe suppression

- D. can produce simultaneous map and weather information
6. For any given circumstances, in order to double the effective range of a primary radar the power output must be increased by a factor of:
- A. 2
 - B. 16
 - C. 4
 - D. 8
7. The prime factor in determining the maximum unambiguous range of a primary radar is the:
- A. pulse recurrence rate
 - B. power output
 - C. size of parabolic receiver aerial
 - D. height of the transmitter above the ground
8. Ignoring pulse length, the maximum pulse repetition frequency (PRF) that can be used by a primary radar facility to detect targets unambiguously to a range of 200 NM is: (pps = pulses per second)
- A. 375 pps
 - B. 782 pps
 - C. 308 pps
 - D. 405 pps
9. The interval in time between the commencement of two consecutive pulses is:
- A. pulse rate
 - B. pulse width
 - C. pulse recurrence frequency
 - D. pulse recurrence period
10. The main advantage of a continuous wave radar over a pulsed radar is:
- A. more complex equipment but better resolution and accuracy

- B. removes the minimum range restriction
- C. smaller more compact equipment
- D. permits measurement of Doppler in addition to improved range and bearing

11.A Primary radar operates on the principle of:

- A. transponder interrogation
- B. pulse technique
- C. phase comparison
- D. continuous wave transmission

12.The speed of a radio wave in nm/sec is:

- A. 300,000
- B. 161,842
- C. 163,842
- D. 186,000

13.A radar facility transmitting at a Pulse Recurrence Frequency (PRF) of 1200 pulses/second will have a maximum unambiguous range of approximately:

- A. 135 NM
- B. 69 NM
- C. 270 NM
- D. 27 NM

14.Which combination of characteristics gives best screen picture in a primary search radar?

- A. Short pulse length and narrow beam
- B. Long pulse length and wide beam
- C. Long pulse length and narrow beam
- D. Short pulse length and wide beam

15. The pulse recurrence frequency of a signal having a pulse interval (pulse recurrence period) of 5 microseconds is:

- A. 6 MHz
- B. 200 KHz
- C. 60 MHz
- D. 2000 KHz

16. What is the maximum theoretical range for a primary radar with a PRF of 324 pps?

- A. 250 nm
- B. 500 nm
- C. 463 nm
- D. 463 sm

17. A primary pulse radar system operates with a pulse repetition frequency of 400 pps. The maximum theoretical range of the system is:

- A. 202 nm
- B. 303 nm
- C. 404 nm
- D. 505 nm

18. Which is the most suitable radar for measuring short ranges:

- A. millimetric pulse
- B. continuous wave primary
- C. centimetric pulse
- D. continuous wave secondary

19. Attenuation of radio waves means:

- A. the weakening of the radiated waves
- B. the atmospheric bending of the waves
- C. only the scattering of the waves by the tropopause
- D. only the absorption of radio energy by the sea

20. In a primary radar system:

- A. the radar is primarily used for range-finding
- B. all radio frequency energy is produced by the radar located at the radar site
- C. the aircraft plays the secondary role, just listening to the radar signals from the ground radar
- D. the radar is the primary aid for ATC

21. In which frequency band do most airborne weather, and ground based ATC, radar systems operate?

- A. SHF
- B. UHF
- C. EHF
- D. VHF

22. The maximum range obtainable from an ATC Long Range Surveillance Radar is approximately:

- A. 100 NM
- B. 200 NM
- C. 300 NM
- D. 400 NM

23. The maximum PRF required for a range of 50 nm is:

- A. 300 pulses per second (pps)
- B. 600 pps
- C. 1620 pps
- D. 3240 pps

24. Complete the following statement. Aircraft Surface movement Radar operates on frequencies in the

- (i) band employing an antenna that rotates at approximately
- (ii) revolutions per minute; it is
- (iii) possible to determine the type of aircraft from the return on the radar screen.

- A. SHF (ii) 10 (iii) always
- B. EHF (ii) 30 (iii) never
- C. SHF (ii) 60 (iii) sometimes
- D. EHF (ii) 100 (iii) never

25. What is the maximum theoretical range, in nm, of a radar whose PRF is 750 pps?

- A. 132 NM
- B. 218 NM
- C. 200 NM
- D. 108 NM

26. What is the range of long range ground radar?

- A. 200 nm
- B. 50 nm
- C. 1000 nm
- D. 300 nm

27. An aerodrome ground movement radar is likely to operate in the _ band with a scan rate of _ revolutions per minute.

- A. EHF, 1000
- B. EHF, 20
- C. UHF, 200
- D. SHF, 60

28. On which of the following radar displays is it possible to get an indication of the shape and to some extent the type, of the aircraft generating the return?

- A. Aerodrome Surveillance (approach) Radar
- B. Airborne Weather Radar (AWR)
- C. Aerodrome Surface Movement Radar (ASMR)
- D. Secondary Surveillance Radar (SSR)

29. What is a typical range for an EN-route surveillance radar (RSR)?

- A. Up to 25 nm
- B. Up to 80 nm
- C. Up to 250 nm
- D. Up to 2.500 nm

30. Which of the following radar equipments operate by means of the pulse technique?

- 1. Aerodrome Surface Movement Radar
- 2. Airborne Weather Radar
- 3. Secondary Surveillance Radar (SSR)
- 4. Aerodrome Surveillance (approach) Radar

- A. 1, 2, 3 and 4
- B. 1, 2 and 4 only
- C. 2, 3 and 4 only
- D. 2 and 4 only

31. Considering a primary radar system, what kind of aeriels are used?

- A. One directional antenna both for transmitting and for receiving
- B. A directional antenna for transmitting, and an omni-directional antenna for receiving
- C. One directional antenna for transmitting and one for receiving
- D. An omni-directional antenna for transmitting, and a directional antenna for receiving

32. When an aircraft is operating its Secondary Surveillance Radar in Mode C an air traffic controller's presentation gives information regarding the aircraft's indicated flight level in increments of:

- A. 200 FT
- B. 100 FT
- C. 250 FT
- D. 150 FT

33. Which one of the following is an advantage of a secondary radar system when compared to a primary radar system?

- A. The relatively small ground antenna transmits no side lobes, thus eliminating the danger of false replies from the airborne transponder
- B. The required power of transmission from the ground equipment is reduced
- C. Possibility of obtaining speed information for aircraft within range
- D. Is not limited to line of sight

34. What is the minimum PRI for a radar with a design range of 200 NM?

- A. 1333 microseconds
- B. 667 microseconds
- C. 1235 microseconds
- D. 2470 microseconds

35. When in tracking mode, the airborne interrogator operates at:

- A. A PRR variable between 24 and 30 pulses per second
- B. A PRR fixed at a rate selected from the range 24 to 30 PPS
- C. A PRR of 150 PPS
- D. A PRR of 2700 PPS

36. Which of the following is a primary radar system?

- A. SSR
- B. DME
- C. GPS
- D. AWR

37. A primary radar has a pulse repetition frequency of 275 pps. The time interval between the leading edges of successive pulse is:

- A. 3.64 milliseconds.
- B. 36.4 milliseconds.
- C. 3.64 microseconds
- D. 36.4 microseconds

38. A primary radar system has a pulse repetition frequency of 450 pps, ignoring pulse width and fly back at the CRT, the maximum range of the radar would be:

- A. 333 nm
- B. 180 nm
- C. 666 nm
- D. 360 nm

39. When flying precision approach radar the vertical displacement is based upon:

- A. Regional QNH
- B. QFE
- C. QFF
- D. Airfield QNH

40. The frequency band and rate of scan of airfield surface movement radars are:

- A. SHF; 60 RPM
- B. SHF; 200RPM
- C. EHF; 100RPM
- D. EHF; 10RPM

41. When carrying out a precision radar approach talk down normally cease at Nm from touchdown

- A. 0.5 nm
- B. 2 nm
- C. 3 nm
- D. 5 nm

42. A ground based radar with a scanner rotation of 60 rpm, a beam width in the order of $.5^\circ$ and a PRF of 10000 pps would be:

- A. An airfield surface movement indicator with a theoretical range of 8nm.
- B. A precision approach radar

- C. An airfield surface movement indicator with a theoretical range of 16nm
- D. A high resolution surveillance approach radar

43. The SHF band has been selected for airfield surface movement indicator (ASMI) radars in preference to the EHF band because:

- A. The EHF band causes unacceptable radiation hazards to personnel
- B. The attenuation caused by precipitation is greater in the EHF band and reduces the radar's effective range and usefulness
- C. The EHF band is not suitable for the provision of the very narrow beams needed for an ASMI radar
- D. Target discrimination using the SHF band is better



SECONDARY SURVEILLANCE RADAR

1. **What most affects the range available from a secondary radar?**
 - A. The transmission power of aircraft interrogator
 - B. The transmission power of ground transponder
 - C. The height of aircraft and height of ground interrogator
 - D. The PRP

2. **The ground Secondary Surveillance Radar (SSR) equipment incorporates a transmitter and receiver respectively operating in the following frequencies (transmitter; receiver):**
 - A. 1090 MHz; 1090 MHz
 - B. 1090 MHz; 1030 MHz
 - C. 1030 MHz; 1090 MHz
 - D. 1030 MHz; 1030 MHz

3. **The frequency of an SSR ground transmission is:**
 - A. 1050 +/- 0.5 Mhz
 - B. 1030 +/- 0.2 Mhz
 - C. 1090 +/- 0.3 Mhz
 - D. 1120 +/- 0.6 Mhz

4. **The ATC transponder system excluding Mode S contains:**
 - A. four modes, each 1024 codes
 - B. two modes, each 4096 codes
 - C. four modes, each 4096 codes
 - D. two modes, each 1024 codes

5. **In order to indicate unlawful interference with the planned operation of the flight, the aircraft Secondary Surveillance Radar (SSR) transponder should be selected to:**
 - A. 7600

- B. 7500
- C. 7700
- D. 7000

6. What is the maximum number of usable Secondary Surveillance Radar (SSR) transponder codes?

- A. 4096
- B. 3600
- C. 1000
- D. 760

7. Which one of the following Secondary Surveillance Radar (SSR) codes should be used by aircraft entering airspace from an area where SSR operation has not been required?

- A. 0000
- B. 5000
- C. 7000
- D. 2000

8. The accuracy of SSR height as displayed to the air traffic controller is:

- A. +/- 25 ft
- B. +/- 50 ft
- C. +/- 75 ft
- D. +/- 100 ft

9. When an aircraft is operating its Secondary Surveillance Radar in Mode C an air traffic controller's presentation gives information regarding the aircraft's indicated flight level that is accurate to within:

- A. + or – 50 ft
- B. + or – 75 ft
- C. + or – 100 ft
- D. + or – 25 ft



10. Why is the effect of returns from storms not a problem with SSR?

- A. The frequency is too high
- B. SSR does not use the echo principle
- C. The PRF is jittered
- D. By the use of MTI to remove stationary and slow moving returns

11. Which SSR mode A code should be selected when entering European airspace from an area where no code has been allocated:

- A. 7000
- B. 7500
- C. 2000
- D. 0000

12. When Mode C is selected on the aircraft SSR transponder the additional information transmitted is:

- A. height based on QFE
- B. altitude based on regional QNH
- C. aircraft height based on sub-scale setting
- D. flight level based on 1013.25 hPa

13. A secondary radar can provide up to 4096 different codes. These 4096 codes can be used in:

- A. mode C only
- B. mode A only
- C. all modes
- D. mode S

14. The availability of 4096 codes in SSR is applicable to mode:

- A. A
- B. C
- C. S
- D. All

15. With regard to the advantages of SSR which of the following statements is correct?

- A. Little power is required to effect longish range
- B. No aircraft manoeuvres are necessary for identification
- C. Range, bearing and height can be calculated from reply signals
- D. All of the above

16. If an aircraft is hijacked it is recommended that the pilot set transponder code:

- A. 7700
- B. 7500
- C. 7600
- D. 7300

17. In order to indicate radio failure the aircraft SSR transponder should be selected to code:

- A. 7000
- B. 7700
- C. 7600
- D. 7500

18. The two main design functions of Secondary Surveillance Radar (SSR) Mode S are:

- A. the elimination of ground to air communications and the introduction of automatic separation between aircraft using TCAS II
- B. collision avoidance using TCAS II and improved long range (HF) communication capability
- C. continuous automatic position reporting using Global Positioning System (GPS) satellites and collision avoidance using TCAS II
- D. air to ground and ground to air data link communications and improved ATC aircraft surveillance capability

19. The code transmitted by a SSR transponder consists of:

- A. phase differences

- B. pulses
- C. frequency differences
- D. amplitude differences

20. Selection of mode C on the SSR provides ATC with information based on:

- A. aircraft height above QFE
- B. aircraft altitude as indicated on the captains altimeter
- C. aircraft pressure altitude
- D. aircraft height above the surface

21. Which statement regarding Mode S transponders is most correct?

- A. Mode S transponders reduce RT traffic and provide a datalink facility
- B. Mode S transponders are used with TCAS III
- C. Mode S transponders are used to assist GPS positioning
- D. Mode S and Mode C transponders operate on different frequencies

22. Why is a secondary radar display screen free of storm clutter?

- A. The principle of echo return is not used in secondary radar
- B. The frequencies employed are too high to give returns from moisture sources
- C. A moving target indicator facility suppresses the display of static or near static returns
- D. The frequencies employed are too low to give returns from moisture sources

23. Which of the following Secondary Surveillance Radar (SSR) codes is used to indicate transponder malfunction?

- A. 7600
- B. 0000
- C. 4096
- D. 9999

24. Which of the following equipment works on the interrogator / transponder principle?

- A. Secondary Surveillance Radar (SSR)
- B. Global Positioning System (GPS)
- C. Airborne Weather Radar (AWR)
- D. Aerodrome Surface Movement Radar

25. Which one of the following switch positions should be used when selecting a code on the Transponder?

- A. NORMAL
- B. OFF
- C. STBY (Standby)
- D. IDENT (Identification)

26. The selection of code 2000 on an aircraft SSR transponder indicates:

- A. unlawful interference with the planned operation of the flight
- B. an emergency
- C. transponder malfunction
- D. entry into airspace from an area where SSR operation has not been required

27. The selection of code 7500 on an aircraft SSR transponder indicates:

- A. unlawful interference with the planned operation of the flight
- B. an emergency
- C. transponder malfunction
- D. radio communication failure

28. The selection of code 7700 on an aircraft SSR transponder indicates:

- A. an emergency
- B. radio communication failure
- C. transponder malfunction
- D. unlawful interference with the planned operation of the flight

29. What are the frequencies used for interrogation and response for SSR?

- A. 1090 MHz for interrogation from the ground 1030 MHz for response from the aircraft
- B. 1030 MHz for interrogation from the ground 1090 MHz for response from the aircraft
- C. 1090 MHz for interrogation from the aircraft 1030 MHz for response from the ground
- D. 1030 MHz for interrogation from the aircraft 1090 MHz for response from the ground

30. With regard to SSR:

- A. The interrogator is on the ground and the transponder is on the ground
- B. The interrogator is on the ground and the transponder is in the aircraft
- C. the interrogator is in the aircraft and the transponder is on the ground
- D. The interrogator is on the aircraft and the transponder is in the aircraft

31. With SSR, interrogation and response signals:

- A. are separated by 63 MHz
- B. must be set by the pilot but are always 60 MHz apart
- C. are at standard frequencies separated by 60 MHz
- D. are at variable frequencies set by the controller but are always 63 MHz apart

32. In the SSR response, the operation of the transponder ident button:

- A. transmits the aeroplanes registration or flight number as a data coded sequence
- B. sends a special pulse after the normal response pulse train
- C. sends a special pulse before the normal response pulse train
- D. sends a special pulse in the X position on the pulse train

33. When a mode C interrogation is responded to, vertical position of the aircraft is coded and transmitted. This vertical position is referred to:

- A. the sub-scale of the altimeter
- B. area QNH

- C. 1013.2 hPa
- D. any of the above as directed by ATC

34.Data transmission and exchange is conducted in:

- A. Mode A
- B. Mode C
- C. Mode D
- D. Mode S

35.Garbling is caused by:

- A. an aeroplane's transponder responding to side lobes or reflections of the interrogation signal
- B. aeroplane is in close proximity responding to the same interrogation
- C. aeroplane at range responding to interrogations from another ATC, SSR
- D. Doppler effect on targets moving radially towards or away from the SSR

36.A mode S transponder will:

- A. not respond to interrogations made on mode A
- B. respond normally to mode A/C interrogations
- C. respond to mode A interrogations but not mode C
- D. not respond to mode A/C as it is on the different frequency

37.What is the maximum number of usable Secondary Surveillance Radar (SSR) transponder codes in Mode A?

- A. 1000
- B. 3600
- C. 4096
- D. 760

38.Which of the following radar equipment operate by means of the pulse technique?

- 1. Aerodrome Surface Movement Radar**
- 2. Airborne Weather Radar**
- 3. Secondary Surveillance Radar (SSR)**
- 4. Aerodrome Surveillance (approach) Radar**

- A. 1, 2 and 4 only
- B. 1, 2, 3 and 4
- C. 2, 3 and 4 only
- D. 2 and 4 only

39. Why do clouds not appear on secondary radar screens:

- A. Too high a frequency
- B. Too low a frequency
- C. They do not provide an echo by returning signals
- D. The transmit and receive signals are on different frequencies

40. If a radar pulse contains 300 cycles of RF energy at a frequency of 600 MHz, the physical length of the pulse is:

- A. 1550 metres
- B. 150 metres
- C. 1.5 metres
- D. 0.15 metres

41. During a surveillance radar approach, the vertical position of the aeroplane, unless advised otherwise, is given in terms of: _____

- A. Height above aerodrome level
- B. Height above area QNH datum
- C. Height above QFE threshold
- D. Height above aerodrome QNH datum

42. In SSR, the interrogations use different modes. If altitude reporting is required, the aeroplane's transponder should be set to "ALT" and will respond to:

- A. Mode C interrogations only
- B. Mode A interrogations only

- C. Mode C and A interrogations
- D. Mode C and Ident interrogations

43. 'Fruiting' is caused by:

- A. Aeroplanes in close proximity responding to the same interrogation
- B. An aeroplane's transponder responding to side lobes or reflections of the interrogation signal
- C. Aeroplane at range responding to interrogations from another ATC, SSR
- D. Doppler effect on targets moving radially towards or away from the SSR

44. A mode A/C transponder will:

- A. Not respond to interrogations made on mode S
- B. Respond to mode S interrogations but cannot send data
- C. Respond to mode S interrogations with limited data
- D. Not respond to mode S as it is on a different frequency

45. Which of the following statements regarding Mode S is most correct:

- A. Mode S is used to assist in GPS calculations
- B. Mode S transponders are used with the radio altimeter
- C. Mode S transponders reduce R/T traffic and also provide the aircraft with a data link facility
- D. Mode S and Mode A/C transponders use different frequencies of operation

46. The vertical position provided by SSR mode C is referenced to:

- A. QNH unless QFE is in use
- B. 1013.25 hPa
- C. QNH
- D. WGS 84 datum

47. The SSR conspicuity code is:

- A. 7000
- B. 2000
- C. 0033
- D. 4321

48. In which of the following meteorological conditions would you expect to encounter an increased distance to the radar horizon:

- A. surface inversion of temperature and humidity
- B. surface steep lapse rate of temperature and humidity
- C. a steep lapse rate of temperature with an inversion of humidity
- D. an inversion of temperature with a steep lapse rate of humidity

49. What information may be displayed on an ATC radar screen connected only to a primary radar system?

- A. Aircraft position only
- B. Aircraft position and SSR code
- C. Aircraft position, SSR code and altitude
- D. Aircraft altitude

50. In ATC surveillance radar procedures, if primary radar fails but coverage continues to be provided by SSR:

- A. Full radar control with standard radar separation will be maintained
- B. Non-radar separation standards will be introduced as soon as possible
- C. All radar assistance will be terminated immediately
- D. Radar assistance will be terminated after standard separation has been introduced

DME

1. **Ground DME responders respond at a frequency:**
 - A. the same as the interrogation signal
 - B. 63 MHz greater than interrogation frequency
 - C. 63 MHz lower than interrogation frequency
 - D. 63 MHz different from interrogation frequency, either above or below

2. **An aircraft at FL 360 is 10 nm plan range from a DME. The DME reading in the aircraft will be:**
 - A. 8 nm
 - B. 11.7 nm
 - C. 10 nm
 - D. 13.6 nm

3. **The aircraft DME receiver is able to accept replies to its own transmission and reject replies to other aircraft interrogations because:**
 - A. transmission frequencies are 63 MHz different for each aircraft
 - B. pulse pairs are amplitude modulated with the aircraft registration
 - C. aircraft interrogation signals and transponder responses are 63 MHz removed from each other
 - D. pulse pairs are discrete to a particular aircraft

4. **The time taken for the transmission of an interrogation pulse by a Distance Measuring Equipment (DME) to travel to the ground transponder and return to the airborne receiver was 2000 micro-second. The slant range from the ground transponder was:**
 - A. 165 NM
 - B. 186 NM
 - C. 296 NM
 - D. 330 NM

5. If a VOR station and a DME station, having different locations, are selected to provide a fix:
- A. two difference IDs will have to be checked
 - B. two positions, being ambiguous, will be presented
 - C. two sets, with separate frequency control, are required in the aircraft
 - D. all 3 answers above are correct
6. What is the maximum distance between VOR and DME/TACAN ground installations if they are to have the same Morse code identifier?
- A. 60 m
 - B. 2000 m
 - C. 600 m
 - D. 300 m
7. When identifying a co-located VOR/DME the following signals are heard in the Morse code every 30 seconds?
- A. 4 identifications in the same tone
 - B. 4 identifications with the DME at a higher tone
 - C. 4 identifications with the DME at a lower tone
 - D. no DME identification, but if the VOR identification is present and a range is indicated then this shows that both are serviceable
8. In which situation will speed indications on an airborne Distance Measuring Equipment (DME) most closely represent the groundspeed of an aircraft flying at FL 400?
- A. When passing abeam the station and within 5 NM of it
 - B. When tracking directly towards the station at a range of 100 NM or more
 - C. When overhead the station, with no change of heading at transit
 - D. When tracking directly away from the station at a range of 10 NM
9. A VOR and DME are co-located. You want to identify the DME by listening to the call sign. Having heard the same call sign 4 times in 30 seconds the:

- A. VOR and DME call signs were the same and broadcast with the same pitch
- B. DME call sign was not transmitted, the distance information is sufficient proof of correct operation
- C. DME call sign is the one with the lower pitch that was broadcast several times
- D. DME call sign is the one with the higher pitch that was broadcast only once

10. A DME in tracking mode subsequently experiences a reduction in signal strength will switch the equipment in the first instance to:

- A. standby mode
- B. search mode
- C. memory mode
- D. signal controlled search

11. The accuracy of a DME:

- A. is approximately ± 0.5 nm
- B. decreases with increase of range
- C. increases with increase of altitude
- D. is approximately ± 2 nm

12. Distance Measuring Equipment (DME) operates in the:

- A. UHF band and is a primary radar system
- B. VHF band and uses the principle of phase comparison
- C. UHF band and is a secondary radar system
- D. SHF band and uses frequency modulation techniques

13. Groundspeed measurement using DME equipment is most accurate flying:

- A. from the station at long range
- B. over the station
- C. towards the station at short range
- D. past the station at short range

14. An aircraft DME receiver does not lock on to its own transmissions reflected from the ground because:

- A. the pulse recurrence rates are varied
- B. DME transmits twin pulses
- C. they are not on the receiver frequency
- D. DME uses the UHF band

15. A DME is located at MSL. An aircraft passing vertically above the station at flight level FL 360 will obtain a DME range of approximately:

- A. 11 NM
- B. 7 NM
- C. 6 NM
- D. 8 NM

16. The DME in an aircraft, cruising at FL 210, fails to achieve lock on a DME at MSL at a range of 210 nm. The reason for this is:

- A. the beacon is saturated
- B. the aircraft is beyond the maximum usable range for DME
- C. the aircraft is beyond line of sight range
- D. the aircraft signal is too weak at that range to trigger a response

17. During a flight at FL 210, a pilot does not receive any DME distance indication from a DME station located approximately 220 NM away. The reason for this is that the:

- A. aeroplane is below the line of sight altitude
- B. aeroplane is circling around the station
- C. altitude is too high
- D. range of a DME system is always less than 200 NM

18. A DME that has difficulty obtaining a lock-on:

(NOTE: PRF = pulse recurrence frequency, PPS = pulses per second)

- A. alternates search mode with periods of memory mode lasting 10 seconds
- B. stays in search mode without a reduction in PRF

- C. stays in search mode but reduces PRF to max. 60 PPS after 100 seconds
- D. stays in search mode but reduces PRF to max. 60 PPS after 15000 pulse pairs have been transmitted

19. On a DME, display counters rotating throughout their range indicates:

- A. ground equipment failure
- B. airborne equipment failure
- C. the airborne receiver is conducting a range search
- D. the airborne equipment is conducting a frequency search

20. The DME (Distance Measuring Equipment) operates within the following frequencies:

- A. 329 to 335 MHz
- B. 962 to 1213 KHz
- C. 962 to 1213 MHz
- D. 108 to 118 MHz

21. Which of the following will give the most accurate calculation of aircraft ground speed?

- A. A VOR station sited on the flight route
- B. A DME station sited across the flight route
- C. A DME station sited on the flight route
- D. An ADF sited on the flight route

22. A typical frequency employed in Distance Measuring Equipment (DME) is:

- A. 100 MHz
- B. 100 GHz
- C. 1000 MHz
- D. 10 MHz

23. For a conventional DME facility Beacon Saturation will occur whenever the number of simultaneous interrogations exceeds:

- A. 80

- B. 100
- C. 200
- D. 60

24. The aircraft DME receiver cannot lock on to interrogation signals reflected from the ground because:

- A. DME pulse recurrence rates are varied
- B. aircraft transmitter and DME ground station are transmitting on different frequencies
- C. reflections are subject to Doppler frequency shift
- D. DME transmits twin pulses

25. The design requirements for DME stipulate that, at a range of 100 NM, the maximum systematic error should not exceed:

- A. + or – 1.5 NM
- B. + or – 3 NM
- C. + or – 0.25 NM
- D. + or – 1.25 NM

26. ICAO specifications are that range errors indicated by Distance Measuring Equipment (DME) should not exceed:

- A. + or – 0.5 NM or 3% of the distance measured whichever is the greater
- B. + or – 1.25 NM plus 0.25% of the distance measured
- C. + or – 0.25 NM plus 3% of the distance measured up to a maximum of 5 NM
- D. + or – 0.25 NM plus 1.25% of the distance measured

27. Height error has the greatest effect on accuracy when an aeroplane is:

- A. over the base line extensions at low altitude
- B. at the base line bisector at low altitudes
- C. at the base line bisector at high altitude
- D. over the base line extension at high altitude

28. In which of the following frequency bands does DME operate:

- A. UHF
- B. SHF
- C. VHF
- D. EHF

29. A DME and VOR have the same ident, are associated and are:

- A. always co-located
- B. within 2,000 feet of each other
- C. within 100 feet of each other
- D. within 600 feet of each other

30. An aircraft passes overhead a DME station at 12000 feet above the station.

At that time, the DME reading will be:

- A. approximately 2 NM
- B. 0 NM
- C. FLAG/OFF the aircraft is within the cone of silence
- D. fluctuating and not significant

31. Regarding the DME system, which one of the following statements is true?

- A. DME operates in the VHF frequency band
- B. The DME measures the phase difference between the reference and variable phase signals to calculate the distance
- C. The transponder reply carrier frequency differs by 63 MHz from that of the interrogation signal
- D. When passing overhead the DME station the DME will indicate 0

32. The operating principle of a DME is the measurement of the:

- A. time between the transmission and reception of radio pulses
- B. frequency change between the emitted wave and reflected wave
- C. frequency of the reflected wave
- D. phase difference between emitted wave and reflected wave

33. When flying at 6000 feet above ground level, the DME indicates 5 NM.

What is the horizontal distance from the aircraft to the overhead DME?

- A. 4.6 NM
- B. 5.2 NM
- C. 4.9 NM
- D. 4.3 NM

34. The DME ground transponder has a PRF of:

- A. 2700 pulses per second
- B. 27 pulses per second
- C. 150 pulses per second
- D. 1500 pulses per second

35. In DME interrogation and responses are separated by 63 MHz in order to:

- A. Avoid station saturation
- B. Differentiate between precision and ordinary DME
- C. Facilitate channel selection
- D. Prevent self-triggering

36. A DME which is listed as operating on an "X" channel is one which:

- A. Is paired exclusively with an ILS frequency
- B. Is paired exclusively with a TACAN frequency
- C. Responds on a frequency 63 MHz higher than the interrogation frequency
- D. Responds on a frequency 63 MHz lower than the interrogation frequency

37. A VOR and DME are frequency paired. The DME identification ends with Z. This denotes that:

- A. The two beacons are co-located
- B. The beacons are supporting the same site but are not co-located
- C. The beacons are greater than 2000 m apart
- D. The beacons are at the same location but are more than 1000 m apart

38. Where a DME uses mechanical counters for displaying the range, continuous rotation of the counters means:

- A. The DME is in a search mode
- B. The DME is in a memory mode
- C. The DME is in a tracking mode due to a turn
- D. None of the above

39. In a certain VORTAC installation the VOR is coding STN and the DME is coding STZ. This means that the distance between the two beacons is in excess of:

- A. 600 m
- B. 100 m
- C. 2000 m
- D. 300 m

40. What happens when a DME in the search mode fails to achieve lock-on:

- A. it stays in the search mode, but reduces to 60 pulse pairs per second (ppps) after 100 seconds
- B. it stays in the search mode, but reduces to 60 ppps after 15000 pulse pairs
- C. it stays in the search mode at 150 ppps
- D. it alternates between search and memory modes every 10 seconds

41. A DME Transceiver does not lock onto its own reflections because:

- A. the PRF of the pulse pairs is jittered
- B. it uses MTI
- C. the interrogation and reply frequencies differ
- D. the reflections will all fall within the fly back period

42. DME uses (i) radar in the (ii) band:

- A. Primary; (ii) UHF
- B. Primary; (ii) SHF
- C. Secondary; (ii) UHF
- D. Secondary; (ii) SHF

43. An aircraft at FL 200 is 220 nm from a DME (0 ft amsl). The aircraft equipment fails to lock on, this is because:

- A. DME is limited to 200 nm
- B. The aircraft is too high to receive the signal
- C. The aircraft is too low to receive the signal
- D. The beacon is saturated

44. A DME transmitter is operating in the search-for-lock phase. Which of the following statements is correct?

- A. the PRF increases, the range counters count down from maximum, the output power is increased
- B. The PRF increases and the range counters count up from 300 to maximum
- C. The PRF increases and the range counters count down from the maximum
- D. The PRF increases and the power is increased

45. If the DME ground transmitter fails

- A. the last measured range is frozen for 20 secs
- B. all range information is immediately lost
- C. the last recorded range is retained until the ground transmitter is restored
- D. the counters continue to rotate for 8 to 10 seconds

46. "DME is a _ radar which provides _ distances between the aircraft and a ground _". Which sequence most accurately completes the above statement?

- A. primary; accurate; transmitter
- B. secondary; earth; transmitter
- C. secondary; slant; transponder
- D. primary; slant; transponder

47. The time from the transmission of the interrogation pulse to the receipt of the reply from the DME ground station is 2000 microseconds (ignore the delay at the DME). The slant range is:

- A. 330 nm
- B. 185 nm
- C. 165 nm
- D. 370 nm

TCAS/ACAS

1. What corrective action is given by TCAS?

- A. Contact ATC
- B. Climbing or descending right turn
- C. Climbing or descending left turn
- D. Climb or descend

2. On EFIS when the RA alert is received, the CDU will display?

- A. Hollow white
- B. Solid white diamond
- C. Solid red square
- D. Solid yellow square

3. On a RA/VSI the eyebrow FLY TO vertical speed indications is shown in which colour?

- A. Green
- B. Gold
- C. Crosshatched white

4. EFIS indications of collision area when RA is cleared?

- A. Green
- B. Audio indications ALL CLEAR ALL CLEAR
- C. White diamond

5. In TCAS if RA is triggered then can it be ignored if?

- A. The pilot is flying VMC
- B. There is no risk of collision
- C. The pilot is under the command of the ATC
- D. All of the above

6. ACAS II gives information to the pilot about collision threats what advice it provides?

- A. It tells the pilot the range of a possible collision threat, but no bearing
- B. It tells the pilot the bearing of a possible collision threat, but no range
- C. It tells the pilot range and bearing of a possible collision threat
- D. It advises the pilot of avoiding action in the vertical plane

7. What is the normal range of an ACAS installation?

- A. 120 nm
- B. 60 nm
- C. 30 nm
- D. 10 nm

8. What information is available from an ACAS I installation?

- A. Range and direction of conflicting traffic
- B. Range, direction and track of conflicting traffic
- C. Direction of conflicting traffic only
- D. Range of conflicting traffic only

9. What indication should a pilot expect on a TCAS I installation if the equipment assesses that another a/c is a probable collision threat?

- A. Resolution advisories are given on a plan display only
- B. Audio and/or visual signals give traffic advisories only
- C. Audio/or visual signals in plan give traffic or resolution advisories
- D. Resolution advisories may be given in the vertical plane

10. How much warning will a TCAS I give a pilot before a possible collision?

- A. 10s
- B. 20s
- C. 30s
- D. 40s

11. Which of the following is true about ACAS II compared with ACAS I

- A. Traffic advisories are given in the vertical plane as well as horizontal
- B. Resolution advisories are given in the vertical plane
- C. Resolution advisories are given in the horizontal plane
- D. Resolution advisories are co-ordinated between all conflicting

12. What indication would a pilot see his ACAS II display if an a/c was assessed as requiring resolution advisory action?

- A. A white square
- B. A blue diamond
- C. A yellow circle
- D. A red square

13. If a pilot sees a solid white diamond on his EFIS, does that tell him?

- A. This is a traffic advisory which will all pass well away from his a/c
- B. This is a traffic advisory which will pass close to his a/c
- C. This is a resolution advisory which the pilot must turn to avoid
- D. This is a resolution advisory which the pilot must change height to avoid

14. What is needed for an a/c's ACAS I equipment to indicate the relative height of a target which is a possible collision risk?

- A. a/c and target must have ACAS I equipment
- B. a/c must have ACAS I equipment, the target must have ACAS II
- C. only the a/c needs ACAS I, the target must have a serviceable mode C
- D. only the a/c needs ACAS I, the target must have a serviceable mode A

15. What is needed for an a/c's ACAS equipment to give horizontal avoiding action from target which is a probable collision risk?

- A. a/c and target must have ACAS III equipment
- B. only the a/c needs ACAS III, the target must have mode S
- C. Only the target needs ACAS III, the a/c can have ACAS II
- D. Only the a/c needs ACAS III, the target can have ACAS II

RADIO ALTIMETER

1. A radio altimeter measures:

- A. True altitude
- B. Pressure altitude
- C. Height above sea level
- D. Height above the ground or water on which the a/c is flying

2. The failure of the radio altimeter cause:

- A. Loss of pressure altitude data
- B. Loss of density altitude data
- C. Loss of altitude data
- D. Loss of height data

3. The radio altimeter reading will be most reliable at

- A. 500'
- B. 1500'
- C. 5000'

4. The radio altimeter suffers from mushing errors

- A. At very low level
- B. At medium heights
- C. At a very great height
- D. When the Tx and Rx aerials are not at the same level

5. The radio altimeter indicates 2000 ft. what is the minimum ht that the a/c could be?

- A. 1935 ft in the cruising altitude
- B. 1935 ft in the landing altitude
- C. 1898 ft in the cruising altitude
- D. 1899 ft in the landing altitude

6. A/C is flying at FL 250 with its altitude alerting system set to that flight level. If enters a climb, at what altimeter indication will the altitude alert warn the pilot?

- A. 25100 ft
- B. 25300 ft
- C. 25500 ft
- D. 25900 ft

7. Radio altimeter determines a/c height by the principle of?

- A. It measures the Doppler shift of a CW transmission
- B. It measure the Doppler shift of a transmission
- C. It measures the time from transmission to reception of CW frequencies
- D. It measures the time from transmission to reception of a pulse signal

8. How does a radio altimeter work?

- A. It measures the Doppler shift of a CW transmission
- B. It measures the Doppler shift of a pulse transmission
- C. It measures the time from transmission to reception of CW frequencies
- D. It measures the time from transmission to reception of a pulse signal

9. What would be a typical wavelength of a radio altimeter transmission?

- A. 7 mm
- B. 7 cm
- C. 14 cm
- D. 1.4 m

10. The radio altimeter indicates 2000 ft. what is the minimum the minimum ht that the a/c could be?

- A. 1935 ft in the cruising attitude
- B. 1935 ft in the landing attitude
- C. 1898 ft in the cruising attitude
- D. 1899 ft in the landing attitude

11. Which of the following information will a radio altimeter give the pilot?

- A. His decision height
- B. His height above the ground ahead
- C. His actual height above the runway he is approaching
- D. All of the above

12. How can radio altimeter information be presented to the pilot?

- A. On a dial with the logarithmic scale
- B. By lights
- C. Aurally
- D. All of the above

13. An a/c flying at FL 250 with its altitude alerting system set to that FL. If it enters a climb, at what altimeter indication will the altitude alert warn the pilot?

- A. 25100 ft
- B. 25300 ft
- C. 25500 ft
- D. 25900 ft

14. Radio altimeters operate in the

- A. VHF band
- B. SHF band
- C. VLF band



TOP CREW AVIATION

15. Radio altimeters operate up to

- A. 2500 ft
- B. 1500 ft
- C. 500 ft

16. The mean frequency on which radio altimeters operate is

- A. 4200 Mhz

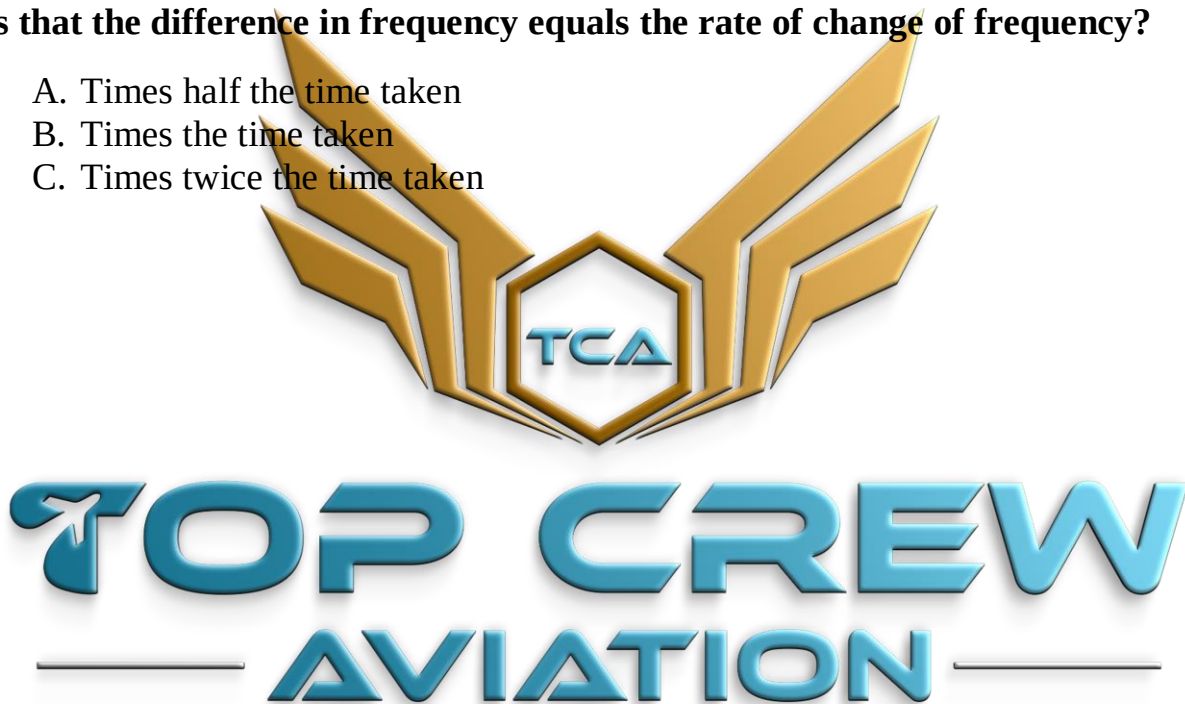
- B. 4300 Mhz
- C. 4440 Mhz

17. A DH setting knob and DH set indication:

- A. Can only be integrated in a radio altimeter in a radio altimeter if it is part of an EFIS
- B. Is usually incorporated in modern radio altimeters
- C. Has to be a separate installation to the radio altimeter itself

18. The formula used to determine the height displayed on the radio altimeter is that the difference in frequency equals the rate of change of frequency?

- A. Times half the time taken
- B. Times the time taken
- C. Times twice the time taken



GROUND PROXIMITY WARNING SYSTEM

1. On EGPWS when in audio alert indications given?

- A. 30 seconds before approaching terrain
- B. 60 seconds before approaching terrain
- C. 60 seconds in closing terrain in red colour
- D. 30 seconds in closing terrain in red colour

2. On RA/VSI the eyebrow FLY TO vertical speed indications is shown in which colour?

- A. Green
- B. Gold
- C. Crosshatched white

3. On a GPWS the mode 5 is And it will be inhibited when mode Is on?

- A. Excessive ROD, mode 1
- B. Excessive ROD, mode 2
- C. Glide slope, mode 3
- D. Glide slope, mode 2

4. An a/c receives a warning on EGPWS. Pilot should:

- A. Disengage autopilot and initiate recovery action
- B. Continue with the autopilot and initiate recovery action
- C. Look around the maneuvering area before attempting any action
- D. Ask ATC for instructions

5. Mark the correct statements in respect of EGPWS?

- A. It gives indications of approaching wind shear
- B. Navigation indications of approaching wind affecting the a/c
- C. Both a and b are correct
- D. Aural warning of wind shear and a visual warning by illuminating a red light

6. EGPWS gives a caution alert:

- A. 60 seconds ahead of terrain conflict
- B. As in a and is repeated every 7 seconds till you are clear of the obstruction
- C. 20 seconds ahead of terrain conflict
- D. 30 seconds ahead of terrain conflict

7. In EGPWS terrain in magenta colour indicates?

- A. Terrain highly above the a/c altitude
- B. Terrain positively below the a/c altitude
- C. Terrain 500 to 2000 ft above the a/c altitude
- D. No database or unknown terrain

8. EGPWS gives warning to the crew

- A. Only if the terrain display is activated
- B. For terrain besides basic GPWS features
- C. Regardless whether the terrain or the obstructions are selected or not
- D. Both b and c are correct

9. EGPWS 50% of terrain indications in red indicates:

- A. Unknown terrain
- B. Obstruction is 1000 to 2000 ft above a/c
- C. Obstruction is more than 2000 ft above a/c
- D. Obstruction threat area warning

10. EGPWS mode 5 glide slope warning below 1000 ft radio altitude comes on when

- A. a/c is 1 dot or more below glide slope
- B. a/c is 1.3 dot or more below glide slope
- C. a/c is 1.5 dot or more below glideslope
- D. a/c is .5 dot or more below glideslope

11. What does a GPWS mode 3 alerts indicate to a pilot?

- A. The a/c is close to the ground with undercarriage up
- B. The a/c is sinking after t/o or go-around
- C. The a/c is below its selected decision height
- D. The a/c is approaching high ground

12. How low on ILS glide slope will the alert the pilot?

- A. Where the a/c is below the bottom of the glide slope
- B. When the a/c is sinking after take-off or go-around
- C. The a/c is below its selected decision height
- D. The a/c is approaching high ground

13. What is the principle of EGPWS?

- A. It combines the radio altimeter with sensors to computer safe descent rates at varying heights
- B. It sends radar signals ahead of the a/c to sense high ground ahead
- C. It gives an infer-red TV picture of the flight path through
- D. It uses GPS and a map database to compute safe clearance from ground ahead

14. Mode 4 on the GPWS gives

- A. Warning of terrain clearance between 50 to 500 ft with LG not down
- B. Warning of terrain clearance between 50 to 500 ft with flaps not down
- C. Both are correct

15. In GPWS the BITE can be achieved?

- A. On ground only
- B. In flight only
- C. Both on the ground and in flight

16. Mode 1 on GPWS gives a warning

- A. Below 2450 ftagl

- B. Below 2450 ftamsl
- C. Of excessive rate of descent with respect to terrain
- D. Both a and c are correct

17. Mode 5 of GPWS, mode 5 is inhibited when modeis in use

- A. Mode 1
- B. Mode 3
- C. Mode 2
- D. Mode 4

18. At what height GPWS starts warning when the flaps are down?

- A. 790 FT
- B. 2450 FT
- C. 500 FT

19. What does a GPWS mode 3 alerts indicated to a pilot?

- A. The a/c is close to the ground with the undercarriage up
- B. The a/c is sinking after t/o or go around
- C. The a/c is below its selected decision height
- D. The a/c is approaching high ground

20. Which of the following would not trigger an alert from the GPWS?

- A. A high sink rate of 1000 ftgda gl in the landing configuration
- B. A high sink rate at 1500 ftagl with gear and flaps up
- C. A low sink rate during go-around
- D. Level flight at 400 ft with undercarriage up

21. How long on the ILS glide slope will the GPWS alert the pilot?

- A. When the a/c is below the bottom of the glide slope
- B. When the a/c is more than 3' below the glide slope
- C. When the ILS indicator shown more than half a full fly down indication
- D. When the ILS indication show more than half a full-up indication

22. Which mode of the GPWS will warn the pilot when he descends below 500 ft with gear still up?

- A. Mode 1
- B. Mode 2b
- C. Mode 3
- D. Mode 4a

23. Below what height will GPWS mode 1 warn the pilot of a high rate of descent?

- A. 2500 ft
- B. 1800 ft
- C. 790 ft
- D. 500 ft

24. Regulations require a pilot to level his wings and initiate a maximum gradient climb to safe altitude under certain conditions. What are these conditions?

- A. Any time he receives a GPWS warning
- B. Any time he receives a GPWS alert
- C. If he receives a GPWS alert, unless in VMC and obviously in no danger
- D. If he receives a GPWS warning unless in VMC and obviously in no danger

25. If GPWS found unserviceable, it must be repaired as soon as possible, If the EBO at destination has got the necessary facilities, what must pilot do?

- A. He must divert to an airfield with repair facilities
- B. He may land at destination, but must wait for the GPWS to be repaired
- C. He may land at the destination, but must make the next landing a repair base
- D. He may continue with his planned schedule until he lands at a repair base

26. How does a pilot know his GPWS is unserviceable?

- A. ATC will tell him
- B. A warning light will illuminate when he descends below 2500 ft
- C. He will test it before flight

D. He has not means of knowing

27. Which of the following circumstances will GPWS not give warning about?

- A. A high rate of descent close to the ground
- B. Rising ground ahead
- C. Approaching with the undercarriage not locked down
- D. Approaching without flaps lowered

28. What is the principle of EGPWS?

- A. It combines the radio altimeter with many sensors to compute safe descent rates at varying heights
- B. It sends radar signals ahead of the a/c to sense high ground ahead
- C. It gives an infra-red TV picture to the flight path through cloud
- D. It used GPS and a map database to compute safe clearance from the ground ahead.



AIRBORNE WEATHER RADAR

1. In which frequency band do most airborne weather radars operate?
 - A. SHF
 - B. UHF
 - C. EHF
 - D. VHF

2. On switching on the AWR a single line appears on the display. This means that:
 - A. the transmitter is unserviceable
 - B. the receiver is unserviceable
 - C. the CRT is not scanning
 - D. the antenna is not scanning

3. In an Airborne Weather Radar that has a colour cathode ray tube (CRT) increasing severity of rain and turbulence is generally shown by a change of colour from:
 - A. green to yellow to red
 - B. yellow to amber to blue
 - C. green to red to black
 - D. yellow to orange to red

4. Which of the following is a complete list of airborne weather radar antenna stabilisation axes?
 - A. Roll, pitch and yaw
 - B. Roll and pitch
 - C. Pitch and yaw
 - D. Roll and yaw

5. Typical Airborne Weather Radar (AWR) is gyro stabilised within limits in:
 - A. pitch and yaw

- B. yaw and turn
- C. pitch and roll
- D. pitch, roll and yaw

6. In which mode of operation does the aircraft weather radar use a cosecant radiation pattern?

- A. MAPPING
- B. CONTOUR
- C. WEATHER
- D. MANUAL

7. If the AWR transmitter is required to be switched on before take-off the scanner should be tilted up with:

- A. either of these modes selected
- B. the mapping mode selected
- C. the weather mode selected
- D. none of these

8. What is the cosecant squared beam of an AWR used for?

- A. To measure cloud height directly
- B. To avoid strong close range ground returns swamping the MAP picture?
- C. To give more detail at range over 60 nm
- D. To remove weather returns from the MAP picture

9. AWR on contour mode what are the position of controls on the AWR panel?

- A. MAP, MAN, WEA, contour
- B. WEA, contour, MAP
- C. MAN, MAP, WEA, contour
- D. Contour, WEA, MAP, MAN

10. AWR operation on the ground is?

- A. Totally prohibited
- B. Unrestrictedly permitted in the aerodrome maintenance areas

- C. Only permitted with certain precautions to safeguard the health of personnel and to protect the equipment

11. AWR works on SHF band because?

- A. Large water drops, hailstones etc good reflection at 3 cm wavelength and small drops like fog, mist, stratus clouds to not give reflection.
- B. It enable the equipment to be powerful but light enough to be installed in the nose cone
- C. It enables narrow beam width to be used to obtain the required degree of definition

12. Narrow conical beam in AWR is required to be used at ranges?

- A. More than 70 nm
- B. Less than 70 nm
- C. At all ranges

13. On an EFIS, AWR picture, what colour depicts the strongest echoes?

- A. Green
- B. Yellow
- C. Blue
- D. Magenta

14. Azimuth resolution on a AWR depends on?

- A. Pulse width of the transmission
- B. Frequency of the transmission
- C. Beam width of the transmission
- D. PRF of the transmission

15. Why is the wavelength of 3 cm used for AWR?

- A. To allow the antenna to fit in the nose of the a/c
- B. To receive reflections from large water drops
- C. To receive reflection from small as well as large clouds
- D. To keep weight of the airborne equipment to a practical level?

16. Which of the following contour indications on AWR would most likely indicate turbulence?

- A. A large red area
- B. A large yellow area
- C. A large green area
- D. Colours changing very rapidly with distance

17. What is the wavelength of typical airborne weather radar?

- A. 10 mm
- B. 30 mm
- C. 10 cm
- D. 30 cm

18. What is the normal maximum range of an AWR displayed on EFIS?

- A. 20 nm
- B. 120 nm
- C. 200 nm
- D. 300 nm

19. An AWR echo at 30 nm when the beam is level with the a/c is lost when the center of the beam is tilted upwards at an angle of 2 deg. How high above the a/c is the top of the rain?

- A. 600 ft
- B. 1500 ft
- C. 3000 ft
- D. 6000 ft

20. Looking at AWR returns on an EFIS screen, where is turbulence most likely?

- A. Where the colour changes rapidly
- B. Where hook-shaped echoes are seen
- C. Both a and b
- D. Turbulence cannot be detected



**TOP CREW
AVIATION**

21. Why is mapping beam of AWR sometimes referred to as the ‘cosecant squared beam’?

- A. Time of scan depends on the cosecant square of the angle from the a/c's nose
- B. Range is found by measuring the cosecant square of the angle of the beam
- C. The cosecant square of the angle of the beam is proportional to the a/c's speed
- D. Transmitter power is proportional to cosecant square of the angle from horizontal

22. By how much should a long range jet pilot avoid strong echoes on his AWR when flying at cruise altitude?

- A. By a minimum of nm
- B. By a minimum of 5 nm
- C. By a minimum of 10 nm
- D. By a minimum of 20 nm

23. What does ‘swept gain’ mean in an AWR?

- A. Transmission power increases as range increases
- B. The returns are stronger at large angles from the a/c heading
- C. Transmission power increases as the cosecant square of the angle from horizontal
- D. The receiver increase sensitivity with time

24. What is an “iso-echo”?

- A. A black indication in the middle of a white radar return
- B. A strong echo showing turbulence
- C. A green indication on EFIS display
- D. A strange shaped return on the AWR screen

25. Flying at FL 270, an AWR weather return at range 40 nm is identifiable when the center of the beam is tilted between +1 deg and – 6 deg. What is the height of base and top of the rain bearing cloud?

- A. Base FL 30, top FL 210

- B. Base FL 30, top FL 290
- C. Base FL 90, top FL 250
- D. Base FL 90, top FL 290

26. Using AWR, the conical beam should be used for map painting in preference to the fan shaped beam when?

- A. These are thunderstorms in the vicinity
- B. Maximum range is required to be scanned
- C. Approaching a coastline

27. AWR operates in the SHF band because?

- A. Large water droplets, hailstones, etc give good reflection at a 3 cm wavelength, while small droplets of fog, mist or stratus do not
- B. It enable equipment to be powerful but lights enough to be installed in the a/c nosecone
- C. It enable a narrow width beam to be used to obtain the required degree of definition

28. The main purpose of selecting HOLD on an AWR is?

- A. The storm movement can be assessed
- B. The storm potential development can be assessed
- C. The storm's potential rate of decay can be assessed

29. Hail associated with thunderstorms will often be shown on an AWR as:

- A. An extremely straight edge to the echo
- B. A finger extending out from the echo
- C. A distinct concave edge to the echo

30. To measure the movement of a thunderstorm when using an AWR, select:

- A. The MOTION switch
- B. The HOLD switch
- C. The TRACK switch

GLOBAL POSITION SYSTEM

- 1. In a Satellite-Assisted Navigation system (GNSS/GPS) a position line is obtained by:**
 - A. the aircraft's receiver measuring the phase angle of the signal received from a satellite in a known position
 - B. timing the period that is taken for a transmission from the aircraft's transmitter/receiver to reach and return from a satellite in a known position
 - C. timing the period that is taken for a satellite's transmission to reach the aircraft's receiver
 - D. the aircraft's receiver measuring the time difference between signals received from a minimum number of satellites
- 2. In which frequency band do Satellite-Assisted Navigation systems (GNSS/GPS) provide position information that is available to civil aircraft?**
 - A. SHF
 - B. EHF
 - C. UHF
 - D. VHF
- 3. What is the minimum number of satellites required by a GPS in order to obtain a three dimensional fix?**
 - A. 3
 - B. 5
 - C. 4
 - D. 6
- 4. Position fixing by satellites is by means of**
 - A. Simultaneous bearing from satellites
 - B. Simultaneous ranges from satellites
 - C. Simultaneous ranges and bearing from satellites

5. How many GPS satellites are needed to provide remote autonomous integrity monitoring for precision guidance?
- A. 3
 - B. 4
 - C. 6
 - D. 3
6. How many GPS satellites are needed to provide an accurate stand alone 3D fix?
- A. 2
 - B. 3
 - C. 4
7. How does a GPS receiver determine from which satellite it is receiving the signal?
- A. Each satellite uses its own frequency
 - B. The signals are sent on the same frequency are coded
 - C. Each satellite sends its signals at different times
 - D. The signals are sent on different frequency and coded
8. Which of the following is not an error of GPS?
- A. Ephemeris error
 - B. Geometric dilution error
 - C. Clock error
 - D. Tropospheric error
9. Which of the following does Dgps not reduce?
- A. Ionospheric group delay
 - B. Multi-path errors
 - C. Selective availability
 - D. Satellite position errors
10. What does a GPS receiver measure?
- A. The time between transmission and reception of a signal from a satellite
 - B. The time for a signal a pass from it to the satellite and return
 - C. The time between reception of signals from 2 satellites
 - D. The phase different between signals from 2 satellites
11. Which GPS signal of code is a available for use by civilians?

- A. The L2 signal
- B. The Y code
- C. The P code
- D. The SPS signal

12. Which of the following is incorrect about the coarse acquisition code of GPS signals?

- A. It is transmitted 90 out of phase from the precision code
- B. It contains the PRN of each individual satellite
- C. It allows an accuracy of +30 m on 99% of occasions
- D. It is encrypted for US military use

13. GLONASS is the Russian equivalent of GPS. Which of the following is correct?

- A. 18 GLONASS satellites cover the world
- B. GLONASS can be combined with GPS to reduce GPS GDOP error
- C. The GLONASS signal is degraded by the Russian military
- D. The GLONASS satellites all transmit on the same frequency

14. Which of the following is incorrect about differential GPS?

- A. It reduces the effect of selective availability
- B. It reduces ionosphere error
- C. It reduces geometric dilution of position
- D. It increases accuracy

15. What does a GPS receiver measure to compute its position?

- A. The range from the satellites
- B. The difference in time from signal transmission to reception
- C. The Doppler difference between the satellite clock and the receiver

16. The Dgps IS?

- A. The time difference between the transmission of the satellites coded
- B. The time difference between the satellite and the a/c GPS until transmitting the same code point on the signal
- C. The correction applied to the 'raw GPS position to obtain a more precise position

17. The reason that the satellite of GPS and GLONASS do not collide is?

- A. The GPS orbit is at 55 deg to the equator while the GLONASS is 60 deg

- B. The GPS satellite orbits are 1200 km higher than the GLONASS orbits
- C. GLONASS has only 3 orbit planes while GPS has 6 orbital planes

18. The GPS transmission frequency which can be used by civil a/c is?

- A. 1575.42 Mhz in the L-band
- B. 2227.5 Mhz in the S-band
- C. 4454.5 Mhz in the C-band

19. The US DoD degrade GPS accuracy?

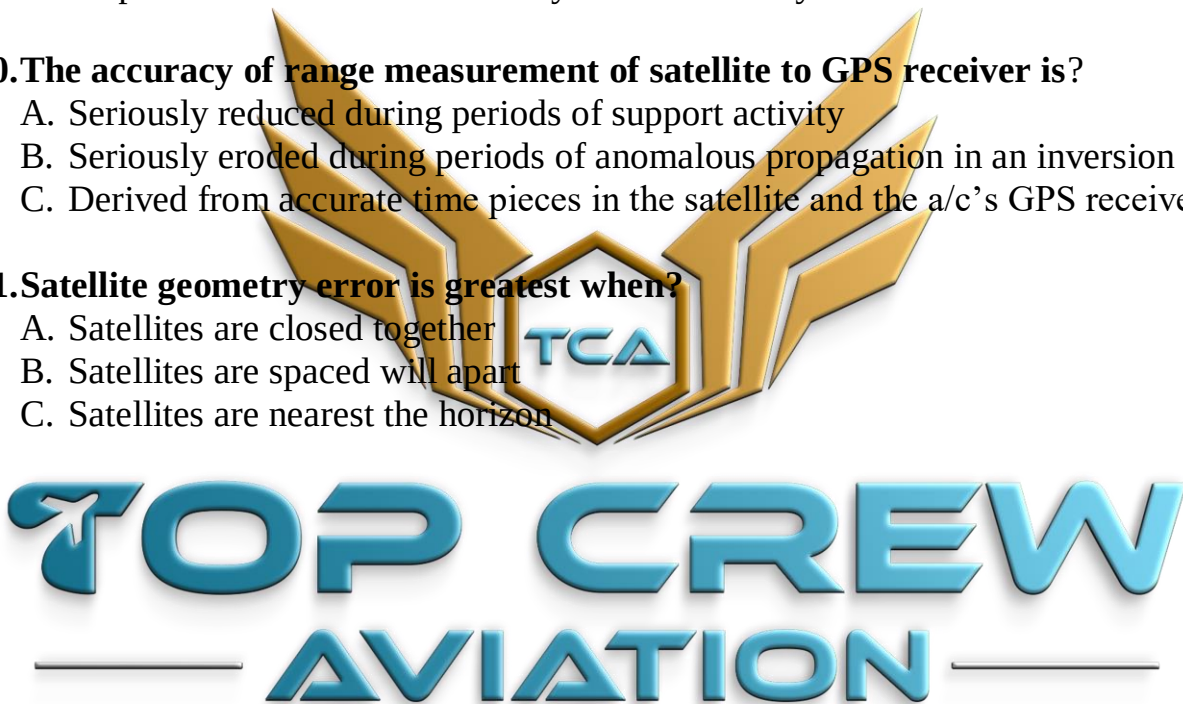
- A. To protect US national security
- B. To protect DoD copyright
- C. To prevent 100% reliance on a system which may have to be

20. The accuracy of range measurement of satellite to GPS receiver is?

- A. Seriously reduced during periods of support activity
- B. Seriously eroded during periods of anomalous propagation in an inversion
- C. Derived from accurate time pieces in the satellite and the a/c's GPS receiver.

21. Satellite geometry error is greatest when?

- A. Satellites are closed together
- B. Satellites are spaced well apart
- C. Satellites are nearest the horizon



RTR(A) & DGCA GROUND CLASSES

THANK YOU!

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